INDUSTRY 4.0 TAILORING

CAPSTONE THESIS

by

Ian Ray Off

Bachelor of Science

Interdisciplinary Studies Engineering

MONTANA STATE UNIVERSITY

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Ian Ray Off

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EXECUTIVE SUMMARY

I created and deployed an Industry 4.0 system to automate the tailoring process.

To scale it, I created a validated business model.

To achieve this, I created an accredited interdisciplinary degree program.

Background

Affordable clothing rarely fits well, because mass manufacturing uses a limited set of non-inclusive templates, while tailoring services are too expensive to regularly use. With modern 3D imaging tools and data processing software, I designed an Industry 4.0 system which uses flexible manufacturing to produce affordable tailored clothing.

To create that system, beyond simple tools and frameworks, I committed to learning and integrating concepts, skills, and ways of thinking from and between several fields. I integrated the departments of computer science, business, industrial engineering, and psychology into a skill set for human factors engineering with smart technology.

Project

Three objectives were established to implement the proposed Industry 4.0 system. Software was created to take measurements from a smartphone-produced 3D image. A fitted garment was produced to demonstrate the viability of software measurements. A business model was field tested and refined to apply the system at industrial scale.

The Exploratory model was used to discover tools and directions for development. The Scrum and Kanban frameworks were used for project oversight and management. Software was created using the Spiral and Rapid Application Development life cycles. A business model was assessed and fine-tuned using the Lean Startup methodology.

Implications

The project results demonstrate that independent data processing software can extract accurate measurements for tailoring from a 3D image captured by a smartphone app. In turn, sets of measurements can be applied to a dynamic template by autonomous flexible manufacturing machines already used by factories for multiple product lines.

This automated tailoring system signifies a potential paradigm shift for the clothing industry, but it is only an early harbinger of the broader fourth industrial revolution. Information technology, smart factories, and AI systems are achieving unprecedented and accelerating advancements which may bring this change to many other industries.

INTRODUCTION

Project Context

This independent capstone project was designed to address the consumer-facing limitations of both traditional clothing manufacturing and clothes-tailoring services. It was inspired by a realization that modern technology such as consumer-grade 3D imaging tools, advanced data processing software, and flexible manufacturing machines could potentially be integrated into a system which automates the tailoring process.

Currently, tailoring is a both time-consuming and very expensive process because it depends on extensive manual labor performed by skilled, trained professionals. Most clothing is instead cheaply mass-manufactured in a predetermined, limited set of sizes and shapes. Therefore, the problem arises that a majority of people cannot or will not pay for tailoring, and many people struggle to find affordable manufactured clothing which fits well. This project explores an alternative solution, where individually fitted clothing can be efficiently mass-produced.

At the time this project was proposed and approved, no prior work had been found regarding automated tailoring systems. Through research and consultation, it was determined that enough of the necessary tools and components existed for the project to have a reasonable chance of success. Flexible manufacturing machines, relatively affordable 3D imaging tools, and 3D data manipulation software all existed, but these components had not been brought together into one process for wearable products.

Although this solution was independently conceptualized, it fits exceptionally well into the broader context of Industry 4.0 (IBM Corp. n.d.). The term Industry 4.0 refers to an ongoing fourth industrial revolution driven by adaptive information technology. By developing smart factories which employ flexible manufacturing systems, it becomes possible to economically achieve ‘mass customization’ in place of traditional, one-size-fits-most mass-production. This project aims to demonstrate the technical design and potential commercial implementation of one such production system for fitted clothing.

As the conceptual scope of this project is far beyond the resources available to implement it, its actual scope is instead that of a minimum viable proof of concept, and the project relied on existing software and hardware wherever possible. The deliverables of this project are as follows: the hardware and software necessary to create a prototype product, the product itself, and a business model using both.

D.I.S. Context

This capstone project was developed in the context of the MSU Bozeman Directed Interdisciplinary Studies degree program (Montana State University n.d.). The following describes its purpose, scope, expectations, and application process:

The D.I.S. degree is for undergraduate students whose scholarly/creative goals fall outside the established degree programs at MSU. It is meant to facilitate unique interdisciplinary skill sets which are valuable in a wide variety of fields including entrepreneurship, leadership, medical school, and law school. A D.I.S. degree is composed of a curriculum which integrates three or more academic disciplines in direct, focused support of a unique research or creative capstone project.

To apply to the D.I.S. program, a student must find and organize a Faculty Advisory Committee (FAC) to represent each discipline. Then, the student must collaborate with their FAC and the D.I.S. director to develop a proposal. This includes the capstone project and approximately 33 precise courses as requirements to complete the degree, representing at least 32 credits in each of three disciplines.

To complete a D.I.S. degree, a student must schedule a thesis defense presentation, and submit their project thesis for review by their FAC and the D.I.S. director. The defense presentation is intended as an opportunity to discuss the capstone project in a flexible public format.

Although some parts of the original proposal are considered set in stone, such as the final deliverables, some flexibility is afforded regarding precise details of the project's execution. It was necessary to develop, propose, and approve the project long before any of the parties involved were certain of whether or how the deliverables could be completed. As such, it was expected that certain adjustments would become necessary over time. Additionally, the course curriculum submitted alongside the original project proposal was continuously reordered and occasionally amended as necessary.

The course schedule for this degree and project includes the following departments:

- Computer Science
- Industrial Engineering
- Business
- Psychology

The Faculty Advisory Committee for this project and their academic disciplines are:

- Dr. John Paxton – Computer Science
- Dr. Frank Marchak – Human Factors Engineering
- Dr. Kregg Aytes – Business / Entrepreneurship

The following documents are:

- The completed course requirements relevant to and necessary for this project
- The original proposal and exact requirements for this project

D.I.S. Curriculum

		Business & Entrepreneurship		
ECNS	101	Economic Way of Thinking	3	
EFIN	101	Introduction to Financial Engineering	1	
M	171	Calculus I	4	
HONR	275	Extreme Microbiology in Yellowstone	3	
BMKT	325	Introduction to Marketing	3	
EGEN	330	Business Fundamentals for Technical Professionals	3	
BMGT	335	Organizational Behavior	3	
EIND	434	Engineering Project Management	3	
BMGT	448	Entrepreneurship	3	
BMGT	461	Small Business Management	3	
PSYX	481	Judgment & Decision Making	3	32
		Human Factors Engineering		
PSYX	110	Introduction to Psychology	4	
EIND	142	Introduction to Systems Engineering	2	
M	172	Calculus II	4	
PHSX	242	Physics II	4	
EIND	313	Work Design & Analysis	3	
PSYX	370	Psychology of Learning	3	
PSYX	375	Behavior Modification	3	
EIND	410	Interaction Design	2	
PSYX	477	Psychology of Wellbeing	3	
CSCI	482	Interdisciplinary Project Instruction	3	
EIND	510	Usability & Inclusive Design	1	32
		Computer Science		
CSCI	112	Programming with C I	3	
CSCI	127	Joy & Beauty of Data	4	
CSCI	132	Basic Data Structures & Algorithms	4	
CSCI	232	Data Structures & Algorithms	4	
M	273	Calculus III Multivariate	4	
M	274	Differential Equations	3	
ESOF	322	Software Engineering	3	
CSCI	347	Data Mining	3	
CSCI	366	Computer Systems	3	
M	386	Software Applications in Mathematics	3	
CSCI	440	Database Systems	3	37

Field	Course	Course Name	Credits

D.I.S. Research/Creative Project Proposal

Degree Title: Software & Technology Entrepreneurship
Project Title: Computer Aided Tailoring

Summary:

The aim of this project is to produce affordable custom-fit clothing, by creating or adapting a 3-D scanning device, 2-D and 3-D modeling software, and a business plan for the commercial sale of such products.

Description:

The inspiration for this project arose from the simple fact that I have trouble finding clothes which fit. For me, this has been a persistent inconvenience for several years, and I balk at the cost of tailoring. This experience has made me particularly aware of similar experiences which others have. Although these inconveniences and discomforts appear in many shapes and forms, I have found that they are surprisingly widespread.

To solve these issues for a wide variety of people, I intend to work toward producing commercially viable, affordable, custom-fit clothing. This project will involve developing a set of tools, a complete product, and a business plan. These tools may change if I discover more effective or cost-effective alternatives. Also, as stages of this project are completed, its scope may expand to the business plan's implementation, and the creation of other tools, products, and applications.

Firstly, I plan to create or adapt a 3D scanning device to take complex bodily measurements. While most pants are at best measured exclusively by inseam length and waist circumference, this tool would allow me to find the perfect fit for anyone by effortlessly taking the equivalent of tens, hundreds, or thousands of measurements. This device could be later adapted to the creation of custom-fit prosthetics, protective gear such as helmets, and computer peripherals such as mice. Additionally, existing products could be measured and stored in a database for online shoppers to use in finding the right size and fit for them.

Secondly, I plan to find or develop 2-D and 3-D modeling software to streamline the application of these measurements. This will be done by casting the measurements from the first stage of this project to a 3-D model of fitted clothing, to be automatically "unraveled" into the 2-D cuts of fabric required to produce it. Although outside of the initial scope, this may later be optimized to minimize waste in the fabric cutting process, and improved by combining measurements from several stances for fitted athletic wear. If necessary, to combine measurements, I am considering the use of narrow artificial intelligence programs.

Thirdly, I plan to produce a proof-of-concept product. This initial product will most likely involve manually sewing together the previously determined cuts of fabric. This product should demonstrably meet both goals of affordability and proper fitting. If possible, this stage may be extended to include research and development for automating the sewing process.

Throughout this process, I plan to develop the framework for a business to produce and sell this product. If possible, this stage may be extended to include starting and managing that business.

Preliminary Work:

So far, I have primarily focused on researching the tools, knowledge, and skills which I will need to implement and complete this project, much of which is displayed in the description of this project. The majority of this preliminary work was devoted to the development of a course plan which I hope will prepare me for every facet of this project, and for its continuation beyond graduation. I believe this course plan will prepare me for further preliminary work which I do not yet have the skills or knowledge to perform.

Deliverables:

This project should result in a business plan for a product (fitted clothing), the product itself, and the software and hardware required to create that product. These deliverables may be expanded upon as described above if time and resources permit. As completing additional stages will increase the viability and profitability of this entrepreneurial venture, I will be motivated to achieve as much as possible before I graduate, while I have access to extensive physical and educational resources.

Potential Sources of Funding:

I currently do not have the skills or knowledge to procure external funding for this type of project. This is part of my motivation for incorporating the discipline of Business/Entrepreneurship into my D.I.S. degree. Whether from classes, mentors, or personal research, I plan to remedy this lack of knowledge, and learn about securing additional funding for my project. Until then, or if no such funding were forthcoming, I expect to personally fund my project by working over summers and during college.

Project Management:

This project is organized into schedule milestones which I have referred to as stages. Each of the four initial stages should be possible to assess through tangible progress, such as a partially or fully functioning device, program, product, or plan. Unfortunately, I am currently unable to create a more comprehensive schedule with solid time-frames for the achievement of these milestones. To resolve this issue, I have developed a course plan which initially focuses on foundational knowledge, prerequisites, and project management skills, and which later includes at least four semesters with time and credits set aside to focus on the project.

Changes Since Proposal

Due to the exploratory nature of this project, the original proposal could not predict or accommodate inevitable changes in context, expectations, and methodology over time. To ensure continuity, each change in direction and expectations since the initial proposal was discussed at length with the D.I.S. program director and relevant members of the Faculty Advisory Committee. The final goal and deliverables of the project were maintained. The following describes what became necessary to change and why, organized by the "stages" described in the original document:

Stage 1

3D imaging software has become a possible viable alternative to the 3D scanning device originally described for the purpose of taking measurements. Both types of tools were thoroughly researched and considered, but the project moved forward with 3D imaging software.

Stage 2

It was determined that original idea of 'unwrapping' a 3D image with modeling software was an unnecessary step in the process of taking and using measurements. Instead, the project moved forward by using data mining techniques to extract measurements from 3D images.

Stage 3

The process and goal of creating a proof of concept product were re-evaluated for efficacy and efficiency. It was decided that the product should represent a simulation of the automated manufacturing, rather than a measure of personal effort and sewing skill. A seamstress was contracted to produce the proof of concept using only the digitally collected measurements which would be automatically fed into a flexible manufacturing machine.

Stage 4

The original step of developing a 'framework for a business to produce and sell this product' was vaguely defined with minimal knowledge of the subject. With the necessary contextual knowledge, this step has been redefined as developing a business model and business model canvas for the automated system.

PROJECT OVERSIGHT

Responsibilities

Due to the complexity of the interdisciplinary program, as well as the tight integration between the project and program, extensive administrative work was incorporated alongside core project work. Core project work was completed during HONR 490 Undergraduate Research/Thesis independent study courses, supplemented by support from open-ended assignments in other courses where possible.

The work responsibilities for this project were organized into the following four categories, in roughly chronological order:

Administration
- Initiate and complete all degree program, course, and project paperwork
- Organize and adjust course schedule to resolve conflicts
- Coordinate with Faculty Advisory Committee and D.I.S. director
- Initiate and organize independent study courses for project work

Research
- Find and evaluate 3D imaging hardware/software
- Find and evaluate 3D data processing methods
- Collect data for business model viability and design
- Independently learn all necessary skills not present in traditional courses

Development
- Test 3D imaging tools and move forward in development with one
- Develop 3D data processing software/tools
- Develop a business model using those tools and a product concept
- Develop a prototype product (fitted clothing)

Defense
- Compile and refine project work and deliverables for final submission
- Submit project for review by Faculty Advisory Committee and others
- Complete graduation paperwork and preparation
- Schedule and attend public project/thesis defense

Methods

This project implemented several project management and lifecycle approaches due to its length, complexity, exploratory nature, and interdisciplinary scope. The following list provides a brief overview of all approaches used:

- The Exploratory model was necessary to develop, organize, and launch the project, then to manage initial research and testing.
- The Scrum framework was necessary to manage a lengthy backlog of requirements in discrete sprints.
- The Spiral model was necessary to manage progressive, extended phases software development.
- The Rapid Application Development model was necessary to manage fast iterative software prototyping within larger phases of development.
- The Lean Startup methodology was necessary to manage iterative testing and then development of a business model canvas.
- The Kanban framework was necessary to manage general short term tasks throughout the project and particularly during the final stretch.

Furthermore, each of these systems addressed a different aspect of the project, and they often overlapped or blended together. For a brief period, all six systems were in use at once. The following sections describe the methods used for general oversight and management of the project as a whole.

Project Development

By necessity, the first part of this project involved the Exploratory Model of software development. This model was generalized from software development to the project as a whole. The Exploratory Model (Rouse 2011) uses the following steps:

1. A project is launched and designed to shed light on a system and expectations.
2. A basic model determined before step 1 is employed and tested.
3. Based on testing, a new model is developed with modifications.
4. The new model is tested for performance and potential future modifications.
5. This testing process is repeated until requirements are met.

This model was used to identify and test potentially viable tools and techniques. Once these were validated, the Exploratory Model was discarded in favor of other, more rigorous, structured, and scalable options for other parts of the project.

Project Operation

General long term aspects of the project were operated using the Scrum framework of project management. Scrum (Sliger 2011) is an implementation of Agile project management which organizes a project into planned backlogs, working sprints, and scheduled review sessions for adaptation. This framework uses the following steps:

1. An organization or customer defines a clear vision of a project
2. If necessary, a 'release planning' meeting is held for long term sprint planning
3. A 'product backlog' is created with all desired results organized by priority
4. The duration of iterations / sprints is established, usually around 1-4 weeks

5. Planning begins for one sprint
6. One estimated sprint-length of items are selected from the product backlog
7. These items are organized into an immutable 'sprint backlog'
8. Work is done according to the sprint backlog with minimal distractions
9. Progress on the sprint backlog is posted to highly visible boards or charts
10. Daily progress, plans, and obstacles are stated in brief daily 'scrum meetings'
11. At the end of a sprint, completed work is demonstrated to stakeholders
12. Progress and plans are discussed with stakeholders in a 'sprint review' meeting
13. The product backlog is updated as necessary

14. Steps 5-13, representing one sprint, are repeated until the project is completed

This framework was used to plan and organize long "sprints" for each semester of academic preparation and project work. This additional management was needed primarily due to the complexity of the D.I.S. curriculum supporting this project.

Normally, course registration is a relatively simple process because degree programs are organized into a recommended sequential progression of courses. This includes roughly 35 credits of required courses, 35 credits of relevant electives, and other core or elective courses, where departmental courses are scheduled to minimize conflicts.

In support of this project, it was necessary to continuously resolve precedence and scheduling conflicts among a backlog of 101 credits of required courses spanning four primary departments and eight others. Using a specialized spreadsheet, each sprint identified a viable set of courses from the backlog, initiated a period of focused work, maintained visual progress monitoring and daily evaluations, and reported results.

Task Management

Short term task management was conducted using a minimal version of the Kanban framework of project management (Radigan 2023). Tasks were visually organized into a long term backlog, a short term to do list, in-progress lists, and a list of completed tasks. As necessary, tasks were further organized by date, priority, or subject.

Milestones

Preliminary Planning

1. Define a project concept
   Done – December 2019
2. Establish initial project specifications
   Done – January 2020
3. Determine necessary tools, knowledge, and skills to learn about
   Done – January 2020
4. Identify and organize a preparatory academic curriculum
   Done – February 2020
5. Complete the project planning and approval process
   Done – February 2020
6. Create tools and procedures to manage ongoing preparation efforts
   Done – March 2020

Preparation

1. Complete introductory courses for all disciplines
   Done – December 2020
2. Develop advanced project management skills
   Done – December 2020
3. Develop general competency in computer science and programming
   Done – December 2020
4. Develop basic interdisciplinary skills necessary for project work
   Done – May 2021
5. Develop advanced and specialized interdisciplinary skills for project work
   Done – December 2021
6. Develop project-essential skills in data mining, design, and business modeling
   Done – May 2022
7. Research and select tools and methods to formally begin project work
   Done – May 2022

Implementation

1. Gain access to hardware and software tools for 3D imaging
   Done – January 2022

2. Create a prototype 3D model and evaluate the selected 3D imaging tools
   Done – May 2022

3. Identify viable directions for further development and testing
   Done – August 2022

4. Select and gain access to new 3D imaging tools
   Done – September 2022

5. Verify effectiveness of 3D imaging tools
   Done – October 2022

6. Select 3D data processing software and techniques
   Done – October 2022

7. Confirm the viability of a data mining pipeline for 3D data processing
   Done – November 2022

8. Extract accurate measurements from a 3D image with automated tools
   Done – November 2022

9. Develop and test a business model using the selected software and techniques
   Done – December 2022

10. Establish a timeline for completion of the project
    Done – December 2022

11. Capture a visually accurate 3D image of a human body
    Done – March 2023

12. Confirm the set of necessary set of measurements to tailor a pair of pants
    Done – April 2023

13. Extract all necessary measurements to complete project deliverables
    Done – April 2023

14. Produce a proof-of-concept prototype product
    Done – April 2023

15. Compile and prepare all project deliverables for evaluation
    Done – April 2023

16. Complete project
    Done – April 2023

END – Introduction

BEGIN – Software Deliverable

SOFTWARE & HARDWARE

Specifications

Architectural Design

Originally, the architectural design of the software deliverable was expected to involve 2D and 3D modeling software. This was proposed as a way to handle and measure object files created with a low-end 3D scanning device. It was thought that a specialized camera for capturing in 3D would produce 3D objects, which would then be stitched together to create a full 3D model of part of a person. Then, the surface of the 3D model would have been split and flattened into the correct 2D template to reproduce an exact copy of the 3D model. It became apparent, during the two years of preparation between the proposal and initial implementation of project deliverables, that this hypothesized process was entirely unrealistic. Fortunately, an alternative approach was found.

A realization occurred that many of the steps in the original process could be bypassed by operating on the 3D data directly as a raw point cloud. A 3D image is captured with a photogrammetry app on a smartphone, then processed directly using data mining techniques. The software deliverable embodies the Data Pipeline architecture, which is designed to cleanse, route, transform, and extract information from data (Stitch n.d.).

Design Patterns

Within the Data Pipeline architecture, the software deliverable implements the Extract-Transform-Load (ETL) design pattern. Data is extracted from a 3D image (data store), then standardized, trimmed, and processed, and finally loaded into another location (Wells 2020). Although this deliverable is only the proof of concept for a larger commercial system, batch processing and data staging areas can be used to optimize ETL systems for large-scale applications.

The following sections address the precise specifications of the software deliverable within the architectural design, and the phrase “the software” refers specifically to the data processing software developed during the course of this project.

Functional Specifications

The software receives input as a 3D image of part of a human body. The software then calculates from that input a set of measurements necessary to produce fitted clothing from a simple template. To achieve this, the software implements the following user-facing and system functionality:

User Actions

- Organize system operations
- Adjust system operations
- Initiate automatic data processing
- Collect processed measurement data

System Operations

- Import 3D image data
- Visualize data
- Orient data for processing
- Position data for processing
- Isolate desired data
- Filter out bad data
- Extract measurements from data
- Report results

Non-Functional Specifications

The software demonstrates that it is possible to collect accurate measurements from with widely available 3D imaging tools. The software is limited in scope to collecting a set of measurements for one garment. However, the code and measurements demonstrate proof of concept for further development. This means that the software is designed to be quickly developed, straightforward, and accurate within its scope.

Interface & Performance Requirements

The software is a proof of concept for an automated commercial data mining pipeline, which would ideally not require any interface or human intervention. Realistically, the high throughput commercial system would require interfaces for monitoring and maintenance. Additionally, machine interfaces may be necessary for other systems managing inputs, outputs, storage, and logistics. Within the scope and resources of this project, a user-facing interface for the software is not viable or necessary.

The performance requirements for the software as a proof of concept are similarly very minimal, although a future commercial version would likely have extensive and complex requirements. Primarily, the response time and throughput of the software are optimized for rapid results and feedback during active development.

Development Standards

The following software development standards were maintained during this project, in roughly decreasing order of importance:

- Prioritize and quickly prototype features to achieve proof of concepts
- Organize development into stages of minimum viable products

- Use existing tools wherever possible to achieve project goals
- Avoid scope creep beyond the requirements of project deliverables.

- Prefer adjustable, extensible, and reusable code to accelerate development
- Prefer automated techniques in accordance with conceptual project goals

- Prefer low cost and open source tools
- Maintain formatted visual representations of operations and outputs

These standards prioritize focused speed and efficiency of software development, in order to ensure successful completion of the prototype product and business model deliverables.

Methods

Software Development

The software deliverable was developed using a combination of two software development life cycles: the Spiral model (Tutorials Point n.d.b) and the Rapid Application Development model (Tutorials Point n.d.a). Two models were selected and integrated because this project introduced a unique situation with one person responsible for multiple roles and scopes of development. Both models and the resulting hybrid system are described by the following steps:

Spiral Model

1. Identify business, system, subsystem, and unit requirements
2. Design conceptual and architectural software specifications for development
3. Construct or extend software to meet the current requirements

    Rapid Application Development Model

    1. Design a 'business model' of requirements and information sources
    2. Analyze, organize, and prioritize data from step 1 into a 'data model'
    3. Define 'process models' to fulfill requirements from step 2 data
    4. Generate prototypes from the step 1-3 models using automation tools
    5. Complete end to end testing of the final prototype and turnover preparation

4. Evaluate progress, costs, and schedule slippage during the iteration, then the feasibility and risks of future work

The Spiral model was used to manage iterative software development and extension at the broad scope of architecture, concepts, and discrete semester-long iterations. This duration was established by two iterations which were conducted as official independent study courses to meet a degree requirement. The first and last steps of the Spiral model were prescribed by the course format, where each iteration built upon the last with new conceptual and architectural specifications. The Spiral model fit well, but something more was necessary to subdivide the roughly fifteen week software construction step into useful pieces for one developer.

At the construction step of each Spiral model iteration, the Rapid Application Development model was used to manage incremental software prototyping on a timescale of weeks, days, or even hours. This model was also useful due to its focus on fast prototyping instead of preplanning, which allowed it to accommodate unknowable initial variables and unexpected challenges in the development process.

Development Tools

The hardware and software tools used during this project to develop the final software deliverable are:

Hardware Tools

- A computer capable of supporting a development environment for 3D data
  - HP Omen 15 Laptop
  - Year: 2020
  - Model: dh1xxx
  - CPU: Intel i7-10750H
  - GPU: Nvidia GeForce GTX 1660 Ti
  - RAM: 32GB (16GB stock)
  - OS: Linux, Fedora 37 Silverblue
- A smartphone capable of installing apps and taking pictures
  - Pixel 7
  - Model: 256GB
  - Year: 2022
  - OS: Android 13
- Rigid and soft measuring tapes to verify software output

Software Development Tools

- Python development environment for *.ipynb Jupyter Notebook files
  - Jupyter-lab server package
  - JupyterLab desktop application
- Support platforms and communities for Python and data science

Software Tools

- Smartphone app for capturing/generating 3D images
  - PolyCam photogrammetry app
- Python libraries for 3D data handling, data mining, and visualization
  - NumPy
  - Matplotlib
  - Scikit-learn
- A tool or library to read and convert 3D image, object, or data files
  - Open3D
- A tool to manually visualize and experiment with 3D data files
  - MeshLab

Initial Testing

The first iteration of software development was organized into several categories of research objectives, each necessary to complete a digital 3D model of a simple object with intact and roughly accurate measurement data. A secondary deliverable, of a mounting solution for a stereoscopic depth camera, was established as a stepping stone in development to either enable or later refine the model creation process. Research and testing in each objective category established how to do the following:

Intel D435i Depth Camera
- Set up and connect a camera, computer, and copy of Intel Realsense software
- Use Intel Realsense software to control a camera and measure image data
- Access camera output directly from OpenCV software
- Export 3D images from a camera in a useful file format (.ply)

Python Programming
- Manage virtual environments to control code dependencies and reproducibility
- Connect virtual environments to a development environment (JupyterLab)
- Use and configure command line interfaces
- Use git for version control and access to open source tools
- Compile tools written in C, C#, or C++ with Python bindings
- Write code to use or integrate existing data processing tools

OpenCV
- Install OpenCV on Windows and Linux
- Read and display image files
- Read color, depth, and other values from multi-channel devices
- Identify possible alternatives to OpenCV

Digital 3D Spatial Reasoning
- Visualize point clouds as collections of 3D numerical data
- Identify opportunities to simplify 3D data processing problems

Point Clouds
- Handle input as raw data, such as nested arrays of numerical color values
- Import and process .ply files using the Point Cloud Library (PCL)
- Apply and use documentation of the internal structure of .ply files
- Use the Point Cloud Library to convert point clouds to other file formats
- Project 3D data into lower dimensions to visualize and simplify operations
- Use linear algebra operations to spatially manipulate 3D data
- Use data mining algorithms to identify clusters in 3D data

3D Modeling
- Use MeshLab for a variety of tasks
- Open, edit, save, and export point cloud files
- Manipulate, translate, and rotate point clouds
- Manually trim regions or clusters of data to isolate 3D data of one object
- Manually align, then register and merge (flatten) multiple point clouds

Camera Mounting Solution

To create a 3D model using the available Intel D435i depth camera, it was necessary to capture, calibrate, and stitch several images from several precise positions and angles around a test object. To that end, extensive technical and market research was conducted to identify potentially viable camera mounts.

First, a rigid platform with multiple mounting points was considered due to its mechanical simplicity, as well as the possibility of cheaply purchasing or constructing it. Through research and experimentation, the following problems were identified:

- Suitable examples of this option were not commercially available
- A manually constructed version would be bulky and difficult to relocate
- Manual adjustment would be required between images
- Multiple synchronized cameras would be required to image unstable objects
- Only one camera was freely available for use in this project
- Additional cameras were too expensive for both testing and future applications
- Image stitching could require more overlap than this option could provide

Second, an articulated camera arm was considered due to its adaptability and known commercial availability in several forms. It was hypothesized that one mobile mounting point could be used to capture images from many different measurable positions and angles. Upon closer evaluation, the following problems were identified:

- Image artifacts in certain positions of the arm itself could complicate stitching
- Extensive manual adjustment would be required between images
- The arm or its mounting point could introduce instability
- Nearly free-floating positioning would require complicated calibration
- Imprecise calibration could necessitate more complicated alignment software
- Advanced alignment software could require excessive object detail

Third, a mounting point on a circular track was considered due to its combination of stability, precision, range of motion, and speed of adjustment. This option was conceptually and mechanically sound, but upon further research, the following problems were identified:

- Full circular track mounts were not commercially available
- The available partial circular track mounts were extremely expensive
- An affordable partial circular track would not provide full functionality
- Constructing a circular track mount would be difficult

Finally, the circular track concept was inverted to that of a turntable for the imaged object itself. This solution to the camera mounting problem allowed images of an object to be precisely captured and calibrated from all horizontal angles. Non-motorized transfer discs and motorized photography stands were identified for imaging a human body. Then, a small, affordable turntable was purchased for testing.

3D Modeling Research

To create the 3D model deliverable, it was necessary to research an exhaustive variety of tools and processes related to 3D imaging and stitching. In roughly chronological order, the following notes were collected about possible tools to use and how they could be used:

OpenCV Stitcher Class

- Homography model transformation
  - For creating photo panoramas
  - Likely not applicable to project but possibly useful
- Affine model transformation
  - For stitching images from specialized devices such as drones and scanners
  - Includes AffineBestOf2NearestMatcher, AffineWarper

Supporting Tools

- OpenCV Rotation Estimator module
  - For image alignment
- OpenCV Group Calib3D
  - For camera calibration to convert between coordinate systems

OpenCV Group Stitching Pipeline

- Steps can be configured, removed, or used separately.
- Likely not applicable to project but possibly useful

OpenCV Structure From Motion Module

- Contains algorithms to reconstruct 3D scenes from 2D images
- Based on a library used in Blender
- Intended for scene reconstruction rather than objects
- Could work if scale is calculated or maintained
- Last of OpenCV point cloud stitching options

Conix Center Pointcloud Stitching

- Multi-camera system for real-time point cloud stitching
- Designed for 10-20 simultaneously connected Intel D400 Series cameras
- Camera calibration uses real world coordinates for each camera
- Calibration requires a Theodolite (physical precision-surveying device)
- Software for large scale commercial applications

Point Cloud Stitching Resources

- MIT-Spark Teaser++
  - Automatic alignment supporting multiple complex transformations
- Point Cloud Library registration module
  - Requires preliminary approximate alignment

Point Cloud Library Tools

- Transformation estimation with rotation
- Singular value decomposition

PCL Iterative Closest Point Pipeline

- Steps to use the pipeline
    - Search for correspondences
    - Reject bad correspondences
    - Estimate a transformation using the good correspondences.
    - Iterate

PCL Feature Based Registration Pipeline

- Could be more complex
- Avoids some data inconsistencies and local minimums during optimization
- Could take advantage of additional non-spatial data.
- Steps to use the pipeline
    - Use SIFT Keypoints
    - Use FPFH descriptors at the keypoints
    - Get the FPFH descriptors and estimate correspondences
    - Use one or many bad correspondence rejection methods
    - Get final transformation

Open Global Registration

- Super4PCS Registration Algorithm
    - Discontinued, replaced by OpenGR fork
- Includes models where other tools focus on environments
- Has tutorials, scripts, PCL integration, and MeshLab integration
- Can use .ply format point clouds
- Must be compiled from source code
- Can register using color as well as spatial data
- Should be able to make use of known rotation values

3D Modeling

- MeshLab Registration
- OpenCV GrabCut Foreground Extraction
- Realsense GrabCut Foreground Extraction

3D Scanning Software

- 3DF Zephyr
- Skanect
- ReconstructMe
- Qlone for Android
- Microsoft 3D Scan for Kinect

Continue Researching

- MIT-Spark Teaser++
- OpenCV Structure From Motion module
- Open Global Registration
- PCL iterative closest point registration
- Simulink point cloud viewer
- General point cloud processing
- General 3D scanning software
- MeshLab for manual point cloud editing

3D Modeling Process

The process of creating the 3D model deliverable was originally organized by recursively dividing the top-level goal into a tree of smaller, prerequisite problems until each one could be solved. Then, each solution was integrated into the solution of a larger problem until the goal was achieved. After the final goal was achieved, the documentation of this recursive process was reorganized into the following linear, bottom-up progression of steps:

Step 1

- Problem
  Point clouds could be defined in an arbitrary coordinate system
- Possibility
  Use a specialized mount to precisely calibrate camera positions
- Solution
  Replace camera mount with rotating stand for a person or object
- Reason
  This greatly simplifies camera calibration placement and alignment by precisely controlling both position and rotation. This could allow stitching algorithms and libraries such as the Point Cloud Library registration module to be used with known rotation values. This could produce more reliable results than an automatic alignment algorithm which can only guess and attempt to optimize transformations.

Step 2

- Problem
  Rough object alignment requires cropping raw point cloud images
- Solution
  MeshLab can be used as a manual proof of concept

Step 3

- Problem
  Precise image alignment for merging requires point cloud registration
- Solution
  MeshLab can be used to achieve this as a manual proof of concept

Step 4

- Problem
  Point cloud images must be stitched to create a 3D model of an object
- Solution
  MeshLab can be used to achieve this as a manual proof of concept

Step 5

- Problem
  Image alignment will need to be automated for future work
- Possibility
  Research and test advanced automatic tools such as MIT-Spark Teaser++

3D Model Deliverable

After 53 iterations of development attempts with the D435i camera, a rough 3D model of an object with roughly accurate measurements was successfully created. However, this avenue of development was then relegated to the position of a last resort option. The originally proposed 3D imaging process proved to be complex, difficult to automate, and not particularly precise.

The camera was mounted on the small tripod included in its original box, and that box was placed on a small turntable in front of the camera. Six images of the box were captured, with 60 degrees of rotation between each image. Each 3D image was exported from the camera as a colored mesh object in .ply file format. Then, many attempts were made to stitch the images together with various software tools.

In the end, to confirm that the images could be stitched together into a viable 3D model at all, it was necessary to manually align and combine the images in MeshLab. The following is an image of the 3D model deliverable from one perspective:

This 3D model had approximately the correct shape and dimensions, but its overall quality was not satisfactory for future development. The primary problem was traced back to the Intel D435i camera, in that alignment was nearly impossible because each image from the camera was low quality and significantly warped.

To prepare for the next iteration of development, a search began for other stereoscopic and LiDAR cameras, as well as potentially cheaper alternative 3D imaging tools.

Prototype Pipeline

The second iteration of software development was organized into six steps and one deliverable with a last resort alternative. The following steps were established before the start of this iteration:

1. Research available technologies to capture point clouds or 3D images
2. Acquire access to or purchase necessary hardware and/or software
3. Learn how to use that technology
4. Use that technology to capture and compile 3D data for an object/person
5. Extract 3D measurements from that data
6. Compile a set of measurements in readable or otherwise usable form

These steps were designed to be nearly impossible to deviate from, but also highly flexible in their individual implementations. The following deliverable or alternative was established as a requirement to complete this iteration:

1. Successful 3D measurements of an object or person

OR

2. Thorough written explanation of what was attempted and why measurements could not be produced with available resources

The first deliverable was the milestone achievement necessary to keep the project on schedule and capable of fulfilling its original scope. The second deliverable was provided as a last resort alternative because it was not certain that a way to complete the first deliverable existed. The first deliverable was completed in full and on time.

The first step identified the PolyCam photogrammetry app as a potentially viable 3D imaging tool. The app was available with a limited number of free 3D image captures, which fulfilled step 2. Brief initial testing of the app completed steps 3 and 4. Step 5 predictably comprised most of the work of this iteration.

To extract measurements from the 3D image of a small test object, a data mining pipeline was created to reformat, orient, position, crop, filter, and calculate distances within the image. This was a prototype of the final pipeline, which needed to calculate more complex measurements from a different shape and size of input data. A 3D image of a Logitech MX Ergo trackball mouse was captured, processed, and measured. This pipeline leveraged tools from the Numpy (Harris 2020), Open3D (Zhou 2018), and scikit-learn (Pedregosa 2011) Python libraries.

The following 14 pages are the prototype pipeline, exported directly from JupyterLab, where it was developed and used. Standard formatting resumes after the exported document.

Prototype Pipeline

April 28, 2023

The following 4 sections import libraries and define functions for the prototype data mining pipeline.

[1]:
```
import open3d as o3d
```

```
Jupyter environment detected. Enabling Open3D WebVisualizer.
[Open3D INFO] WebRTC GUI backend enabled.
[Open3D INFO] WebRTCWindowSystem: HTTP handshake server disabled.
```

[2]:
```
import numpy as np
from matplotlib import pyplot as plt
from sklearn.decomposition import PCA
from sklearn.cluster import DBSCAN
```

The Visualize function renders a specified 2D projection of a point cloud, colored by Z value (vertical) for visual clarity. This is used to review the data from different angles after manipulation.

[3]:
```
def visualize(data, axNum1, axNum2):
    ax1 = data[axNum1]
    ax2 = data[axNum2]

    fig = plt.figure(figsize=(5, 5), dpi=100)
    ax = fig.add_subplot(111)
    ax.scatter(ax1, ax2, c=data[2], alpha=1)
    ax.set_xlim([-0.2, 0.2])
    ax.set_ylim([-0.2, 0.2])
    plt.show()
```

The checkMeasures function prints the actual measurements of the test object, then the current dimensions of a point cloud of the object, in order to evaluate each stage of processing.

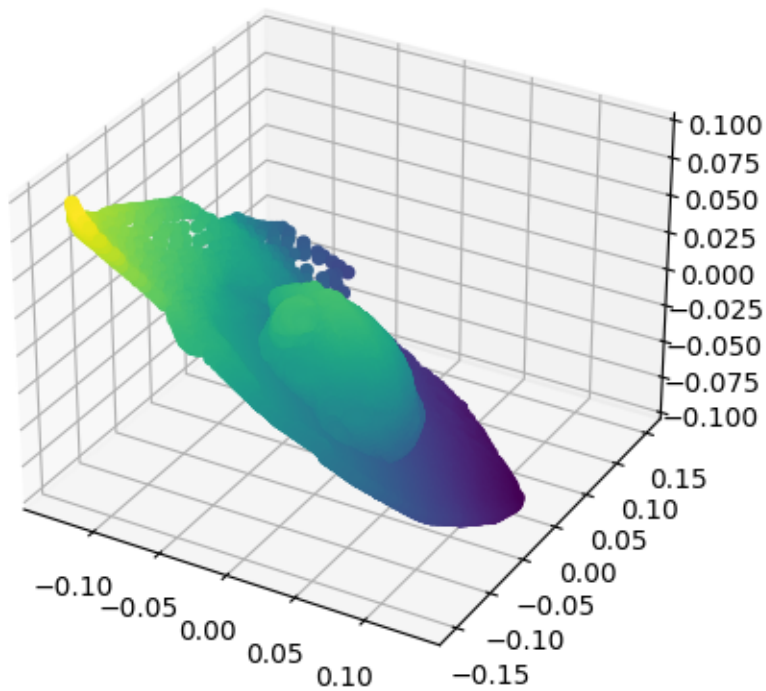
[4]:
```
def checkMeasures(data):
    print("Real measurements: Length 133 mm; Height 60 mm; Width 100 mm")

    print("Length is: {} mm".format(round((abs(np.min(data[0])) + abs(np.
↪max(data[0]))) * 1000)))
    print("Width is: {} mm".format(round((abs(np.min(data[1])) + abs(np.
↪max(data[1]))) * 1000)))
    print("Height is: {} mm".format(round((abs(np.min(data[2])) + abs(np.
↪max(data[2]))) * 1000)))
```

The following section begins the data mining pipeline by importing A 3D image from a file. The image was captured with the PolyCam photogrammetry app, then exported to one of many available file formats with no preprocessing or alterations. As the .ply file format contains a great deal of additional data, Open3D was used to extract only the 3D coordinates of each point/vertex.

[5]:
```
cloud = o3d.io.read_point_cloud('mouse.ply')
```

The following section provides the initial rough visualization of the 3D input data, colored by Z value (vertical) for visual clarity.

[6]:
```
npCloud = np.asarray(cloud.points).T

fig = plt.figure(figsize=(5, 5), dpi=100)
ax = fig.add_subplot(111, projection='3d')
ax.scatter(npCloud[0], npCloud[1], npCloud[2], c=npCloud[2], alpha=1)

plt.show()
```



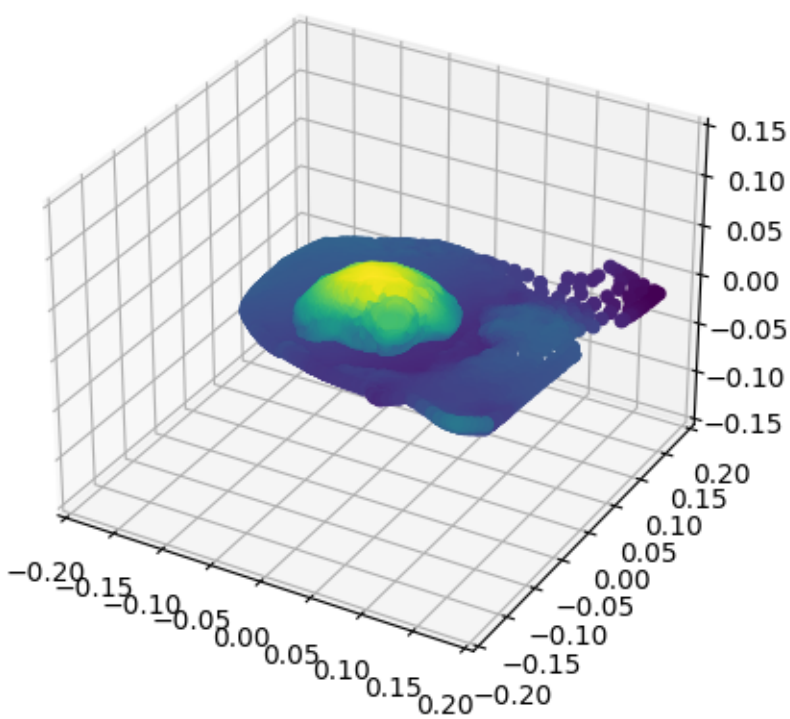
The following section prints an abbreviated copy of the input data and its total size in order to understand the current structure and formatting.

[7]:
```
print(npCloud)
print(npCloud.size)
```

```
[[ 0.08124384  0.06018527  0.05978389 …  0.01187417 -0.0150117
   0.01745604]
 [-0.07352097  0.01447199  0.0095045  … -0.06863514 -0.06274234
  -0.06613733]
 [-0.03189183 -0.0660158  -0.06487438 …  0.03162692  0.02053536
   0.03308232]]
83784
```

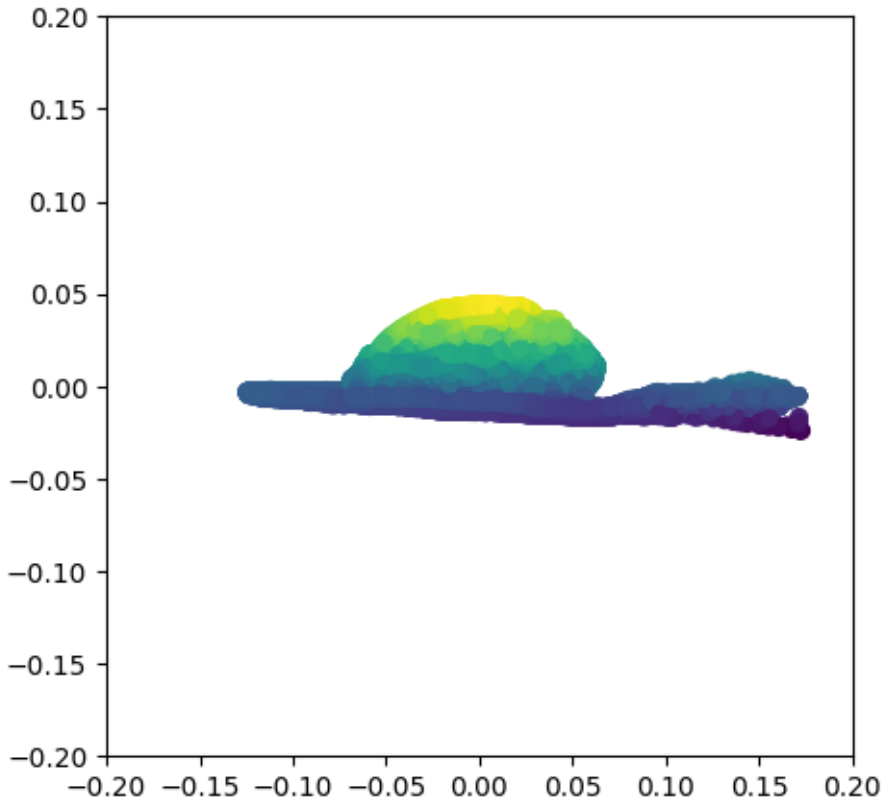
After several other attempts, principal component analysis (PCA) was used to center the input data around the origin (0, 0, 0) and align it with each axis. This method calculates the axis of greatest variance, then the next perpendicular axis of greatest variance, then the next, and so on for higher-dimensional data. Then, the data is projected onto these new axes. Depending on the shape of the input data, it may be necessary to reorder the new axes. If the approximate shape of a 3D image cannot be predicted, further processing may be necessary to determine the correct orientation for each image.

[8]:
```
pca3 = PCA(n_components=3)
pcaData = pca3.fit_transform(npCloud.T)

pcaT = pcaData.T

x = pcaT[0]
y = pcaT[1]
z = pcaT[2]

fig = plt.figure(figsize=(5, 5), dpi=100)
ax = fig.add_subplot(111, projection='3d')
ax.scatter(x, y, z, c=z, alpha=1)
ax.set_xlim([-0.2, 0.2])
ax.set_ylim([-0.2, 0.2])
ax.set_zlim([-0.15, 0.15])
plt.show()
```



The following section renders the X and Z axes of the transposed PCA data in order to confirm that the object is upright, and that the surface under it is close enough to level for further processing.

[9]:
```
visualize(pcaT, 0, 2)
```



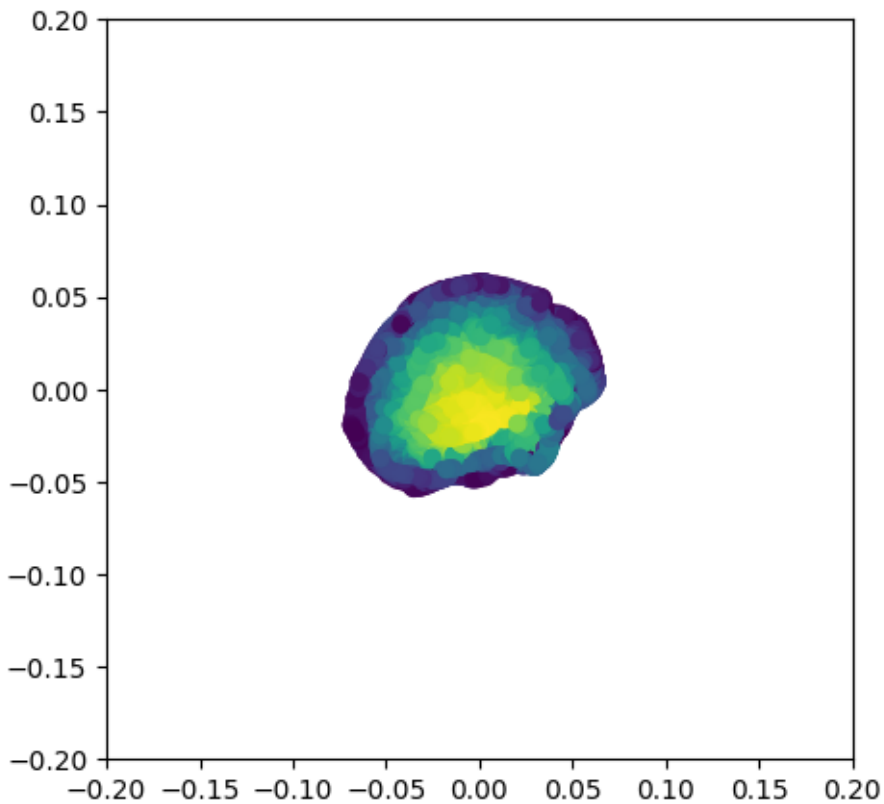
The following section compares the dimensions of the aligned but unfiltered data against the actual measurements of the object.

[10]:
```
checkMeasures(pcaT)
```

```
Real measurements: Length 133 mm; Height 60 mm; Width 100 mm
Length is: 298 mm
Width is: 333 mm
Height is: 68 mm
```

The following section attempts to isolate the object by determining the highest Z value among the four points with the highest and lowest X and Y values, then filtering out all data at or below that Z value.

[11]:
```
minx = pcaData[np.argmin(pcaT[0])]
miny = pcaData[np.argmin(pcaT[1])]
maxx = pcaData[np.argmax(pcaT[0])]
maxy = pcaData[np.argmax(pcaT[1])]
minmax = np.array([minx, miny, maxx, maxy])

cut1 = pcaData[pcaT[2] > np.max(minmax.T[2])].T
visualize(cut1, 0, 1)
```



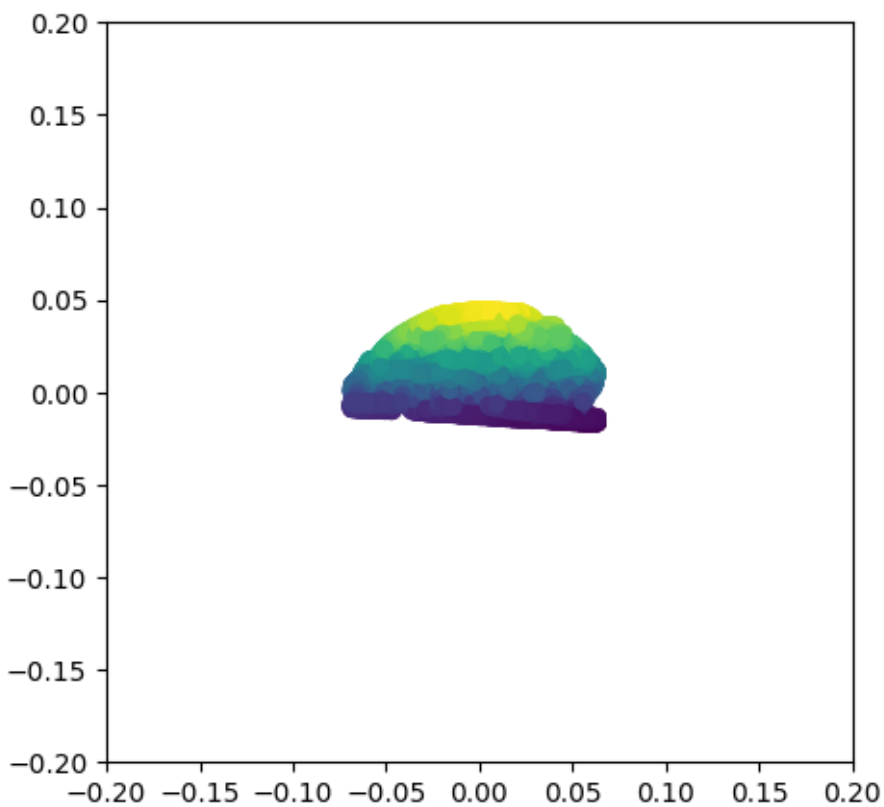
The following section tests the dimensions of the remaining data against the actual measurements of the object, and shows that part of the object was removed.

[12]:
```
checkMeasures(cut1)
```

```
Real measurements: Length 133 mm; Height 60 mm; Width 100 mm
Length is: 132 mm
Width is: 112 mm
Height is: 48 mm
```

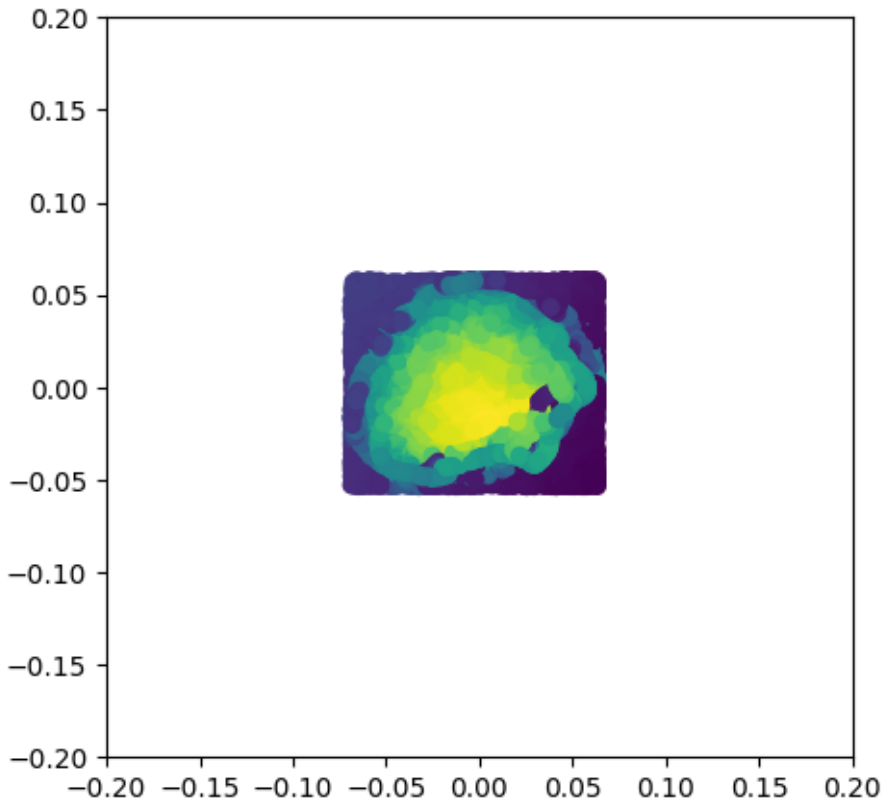
The following section extracts a different portion of the aligned data based on the approximate length and width of the object obtained in the first extraction attempt, then renders a vertical projection of the result.

[13]:
```
cut1minx = np.min(cut1[0])
cut1miny = np.min(cut1[1])

cut1maxx = np.max(cut1[0])
cut1maxy = np.max(cut1[1])

cut2 = pcaData[np.logical_and(np.logical_and(pcaT[0] < cut1maxx, pcaT[1] <␣
↪cut1maxy), np.logical_and(pcaT[0] > cut1minx, pcaT[1] > cut1miny))].T

visualize(cut2, 0, 2)
```



The following section renders a horizontal projection of the same data displayed by the previous section.

[14]:
```
visualize(cut2, 0, 1)
```



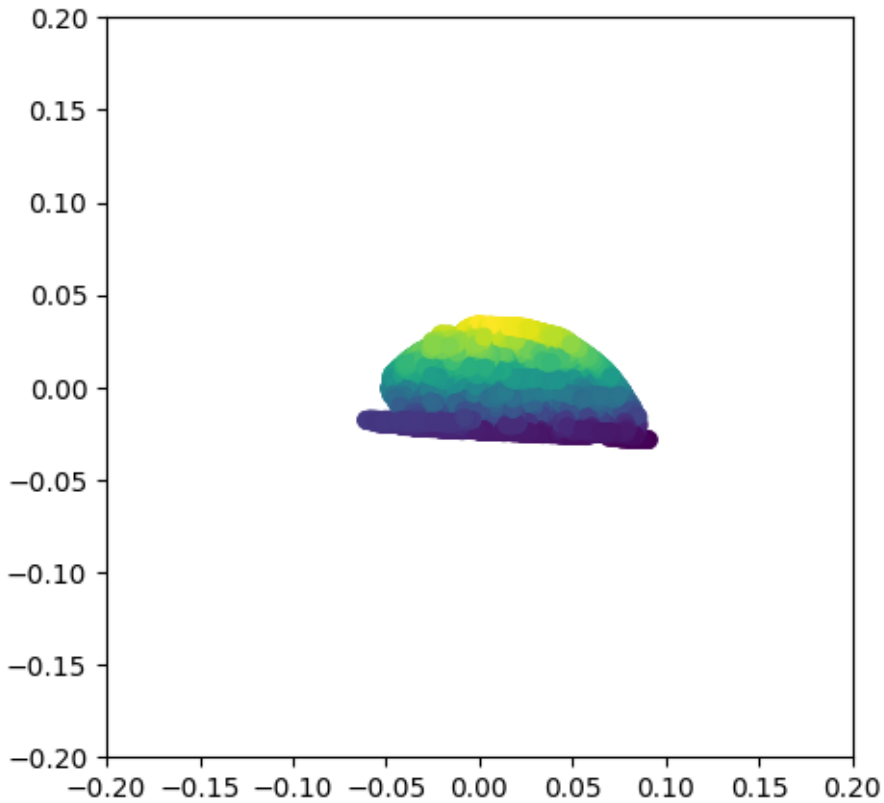
The following section compares the dimensions of the extracted data against the actual measurements of the object in order to evaluate progress.

[15]:
```
checkMeasures(cut2)
```

```
Real measurements: Length 133 mm; Height 60 mm; Width 100 mm
Length is: 132 mm
Width is: 112 mm
Height is: 61 mm
```

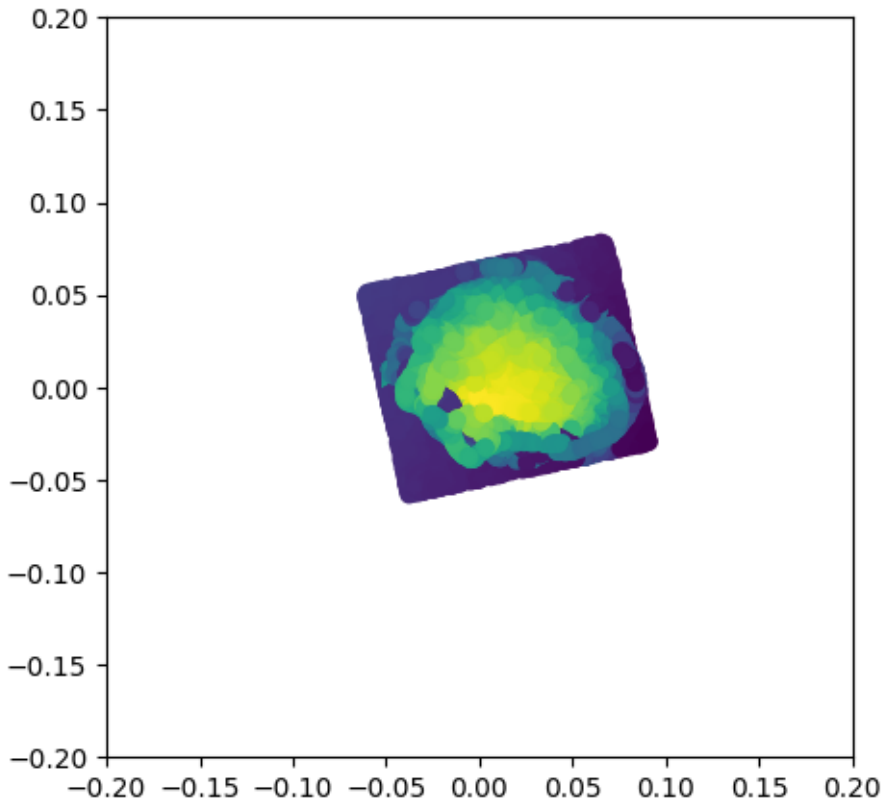
The following section uses PCA again to realign the extracted data for continued processing, and renders a vertical projection of the result.

[16]:
```
cut2pca = pca3.fit_transform(cut2.T).T

visualize(cut2pca, 0, 2)
```



The following section renders a horizontal projection of the same data displayed by the previous section

[17]:
```
visualize(cut2pca, 0, 1)
```



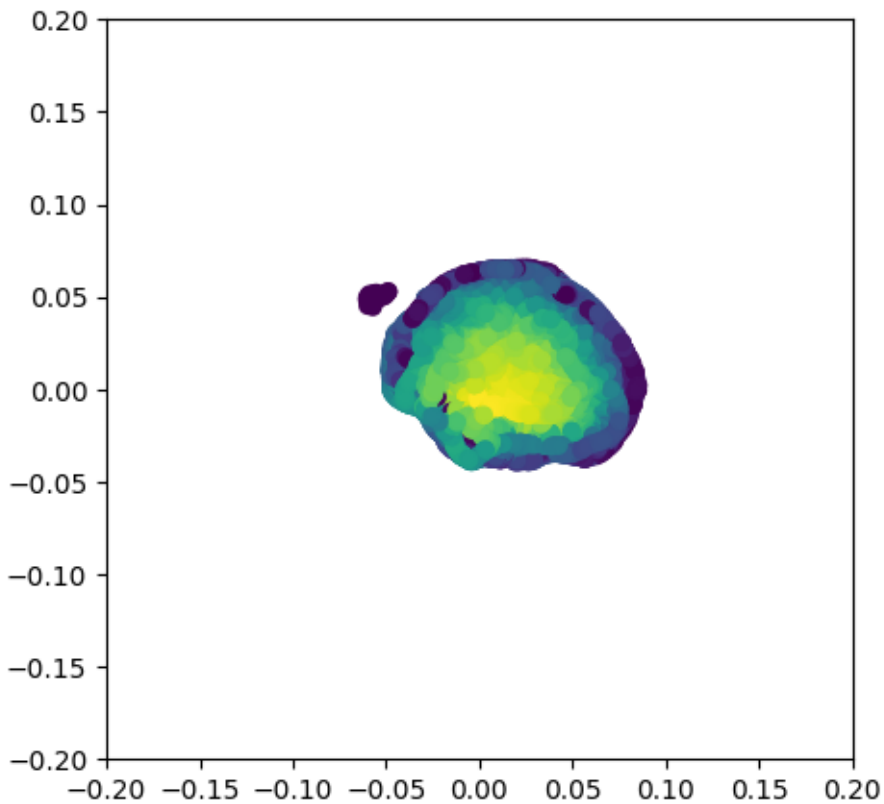
The following section compares the dimensions of the realigned data against the actual measurements of the object in order to determine the effects of realignment.

[18]:
```
checkMeasures(cut2pca)
```

```
Real measurements: Length 133 mm; Height 60 mm; Width 100 mm
Length is: 152 mm
Width is: 136 mm
Height is: 62 mm
```

The following section repeats the process from the first extraction attempt, but with much tighter tolerances due to the other processing steps. The highest Z value is found among the four points with the highest and lowest X and Y values. Then, all data at or below that Z value is filtered out.

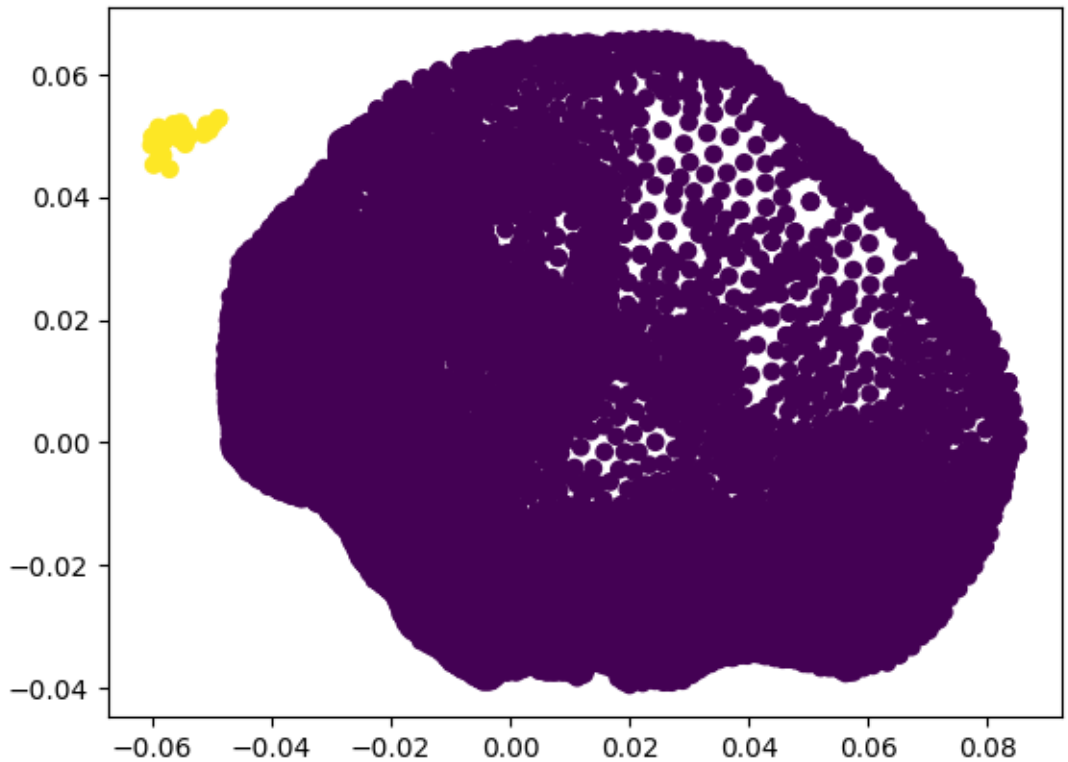
[19]:
```
minx = cut2pca.T[np.argmin(cut2pca[0])]
miny = cut2pca.T[np.argmin(cut2pca[1])]
maxx = cut2pca.T[np.argmax(cut2pca[0])]
maxy = cut2pca.T[np.argmax(cut2pca[1])]
cut2pcaminmax = np.array([minx, miny, maxx, maxy])

cut3 = cut2pca.T[cut2pca[2] > np.max(cut2pcaminmax.T[2])].T
visualize(cut3, 0, 1)
```



The following section uses the DBSCAN clustering algorithm to identify the desired object and extraneous unfiltered data from its surroundings, then visualize the identified data clusters.

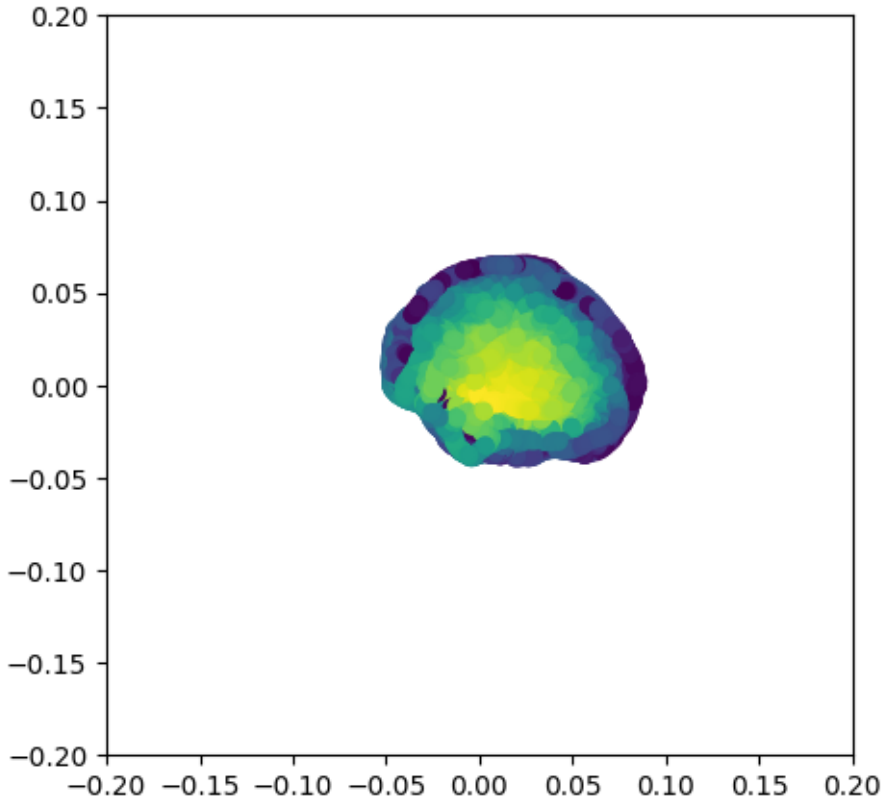
[20]:
```
dbs = DBSCAN(eps=0.01, min_samples=20)
pred_labels = dbs.fit_predict(cut3.T)

plt.scatter(cut3[0], cut3[1], c=pred_labels)
```

[20]:
```
<matplotlib.collections.PathCollection at 0x7f867856e4a0>
```



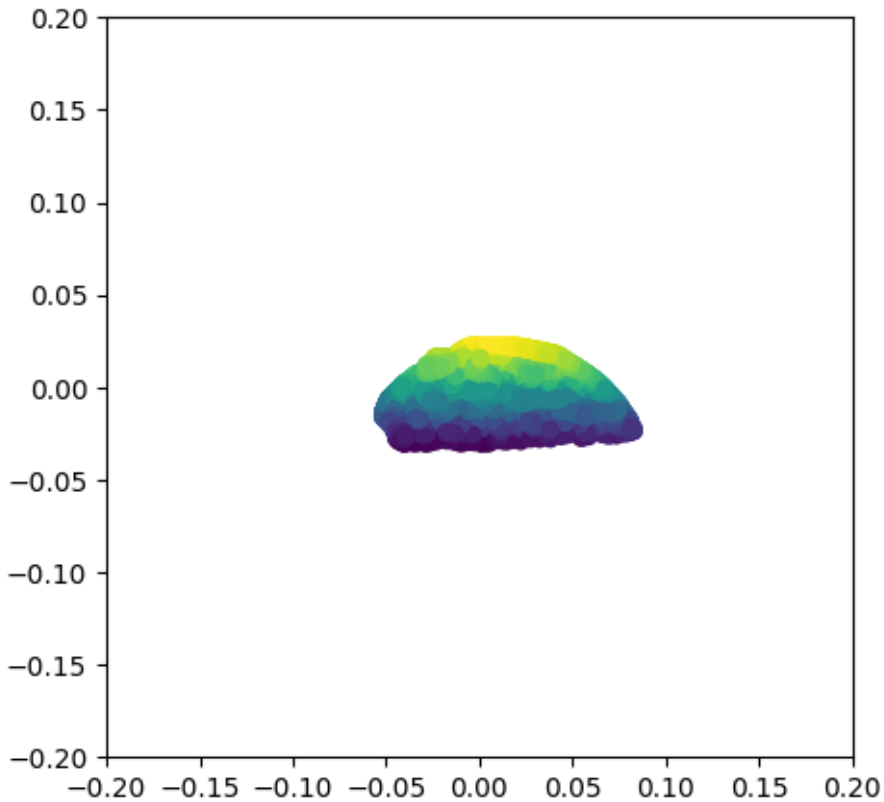
The following section extracts the first and largest cluster of data from the previous section, and renders the result in order to confirm the successful isolation of the desired object.

[21]:
```
cut4 = cut3.T[dbs.labels_ == 0].T
visualize(cut4, 0, 1)
```



The following section realigns and then reorients the isolated object data for measurement.

[22]:
```
cut4pca = pca3.fit_transform(cut4.T).T
cut5 = cut4pca[0], cut4pca[1], -cut4pca[2]

visualize(cut5, 0, 2)
```



The following section checks the measurements of the fully isolated and processed object data against the measurements of the actual object. These final measurements achieved the targeted level of precision, and demonstrated viability for continued development with similar methods.

[23]:
```
print("Point cloud version 5 - final")
checkMeasures(cut5)
```

```
Point cloud version 5 - final
Real measurements: Length 133 mm; Height 60 mm; Width 100 mm
Length is: 134 mm
Width is: 109 mm
Height is: 53 mm
```

RESULTS – Software Deliverable

The third and final iteration of software development was informally organized into five stages. The following objectives marked the completion of these stages:

1. Generalize 3D imaging process from a small object to a human body
2. Identify or implement a way to preprocess the image and isolate the body
3. Identify necessary measurements for tailoring within the image data
4. Create software tools to isolate and calculate all necessary measurements
5. Calculate, test, and optimize measurements while minimizing human input

These objectives encompassed all remaining milestones to complete the software deliverable, with the prototype pipeline from the previous iteration as a launching point. The software deliverable leveraged tools from the Numpy (Harris 2020), Open3D (Zhou 2018), and scikit-learn (Pedregosa 2011) Python libraries.

The PolyCam app was used again for 3D imaging, but also to evaluate the viability of user-accessible 3D image editing tools as an alternative preprocessing method. Although several alternatives to PolyCam were identified during early research, the prototype pipeline represented the only known tool for its purpose. Therefore, the in-app editing tools were an important opportunity to confirm the existence of another viable option with different strengths and weaknesses.

Capturing an accurate 3D image of the lower half of a human body proved to be a very different challenge from imaging a small rigid object with distinct edges. After several different experiments and brief avenues of research, a solution was found to optimize the photogrammetry process. To reconstruct an accurate 3D image, it was necessary to maximize the uniqueness of 2D image inputs from different angles while maintaining a continuous pattern to fit them together. Therefore, imaging was done while wearing a pair of pants with colorful and irregular floral patterns on black fabric.

The following 16 pages are the final software deliverable of the project, exported directly from JupyterLab, where it was developed and used. Standard formatting resumes after the exported document.

Software Deliverable

April 28, 2023

The following seven sections import libraries and define functions to be used throughout a data mining pipeline.

[1]:
```
import open3d as o3d
```

```
Jupyter environment detected. Enabling Open3D WebVisualizer.
[Open3D INFO] WebRTC GUI backend enabled.
[Open3D INFO] WebRTCWindowSystem: HTTP handshake server disabled.
```

[2]:
```
import numpy as np
from matplotlib import pyplot as plt
from sklearn.decomposition import PCA

np.set_printoptions(suppress=True)
```

The checkMeasures function prints the maximum dimensions of a point cloud passed to it. This is used for manual testing to confirm that the input point cloud is about the size it should be.

[3]:
```
def checkMeasures(data):
    print("Width is: {} mm".format(round((abs(np.min(data[0])) + abs(np.
 ↪max(data[0]))) * 1000)))
    print("Length is: {} mm".format(round((abs(np.min(data[1])) + abs(np.
 ↪max(data[1]))) * 1000)))
    print("Height is: {} mm".format(round((abs(np.min(data[2])) + abs(np.
 ↪max(data[2]))) * 1000)))
```

The flatVisualize function renders a 2D scatter plot of input data which is a cross section of a prepared point cloud. This is used to observe and determine how to filter patterns of good or bad data during manual testing.

[4]:
```
def flatVisualize(data):
    xAx = data[0]
    yAx = data[1]

    fig = plt.figure(figsize=(3, 3), dpi=100)
    ax = fig.add_subplot(111)
    ax.scatter(xAx, yAx)
    plt.show()
```

The circumference function calculates the approximate circumference (polygonal perimeter) of a set of (ideally many) 2D input points, and filters out bad data by comparing certain ratios. Where the previous functions are primarily manual diagnostic tools, this function is more involved because it is the key to the most complicated and difficult measurements necessary to precisely fit clothing.

[5]:
```
def circumference(csPts):

# Filter for cross sections with too few points for a correctly shaped perimeter
    if (len(csPts) < 6):
        return 0

# Arrays of True or False for x or y values being positive or negative
    xp = csPts[:, 0] > 0
    xn = csPts[:, 0] < 0

    yp = csPts[:, 1] > 0
    yn = csPts[:, 1] < 0

# Arrays of points in graphical quadrant 1, 2, 3, or 4
    q1 = csPts[np.logical_and(xp, yp), :2]
    q2 = csPts[np.logical_and(xp, yn), :2]
    q3 = csPts[np.logical_and(xn, yn), :2]
    q4 = csPts[np.logical_and(xn, yp), :2]

# Points in each quadrant sorted by X values in clockwise order
    q1s = q1[q1[:, 0].argsort()]
    q2s = np.flip(q2[q2[:, 0].argsort()], axis=0)
    q3s = np.flip(q3[q3[:, 0].argsort()], axis=0)
    q4s = q4[q4[:, 0].argsort()]

# Clockwise-sorted cross section points, for conceptual reference
    csSort  = np.concatenate([q1s, q2s, q3s, q4s])

# Sorted cross section points for the left and right halves of the body
    csSort1 = np.concatenate([q1s, q2s])
    csSort2 = np.concatenate([q3s, q4s])

# Initialized values
    sCirc1 = 0
    sCirc2 = 0
    xMax1 = 0
    xMax2 = 0
    circ = 0

# Matrix operations to calculate the distance between all right side points
# These ensure that there is at least one input point to operate on
    if (len(csSort1) > 0):
```

```
# Appends a duplicate of the final point to the end of csSort1A
# Increases length by one for compatibility but maintains initial positions
        csSort1A = np.concatenate([csSort1, [csSort1[-1]]])

# Appends a duplicate of the first point to the start of csSort1B
# Increases length to cycle all points one position forward
        csSort1B = np.concatenate([[csSort1[0]], csSort1])

# Subtracts 1A from 1B, producing all x and y distances between points
# Uses pythagorean theorem to calculate distances between all points
        sCirc1 = sum(np.sqrt(np.sum(np.square(csSort1B - csSort1A), axis=1)))

# Calculates the absolute value of the maximum X value on the right side for␣
 ↪comparison
        xMax1 = abs(np.max(csSort1A.T[0]))





# Matrix operations to calculate distance between all left side points
# Otherwise functions identically to the previous if statement
    if (len(csSort2) > 0):
        csSort2A = np.concatenate([csSort2, [csSort2[-1]]])
        csSort2B = np.concatenate([[csSort2[0]], csSort2])
        sCirc2 = sum(np.sqrt(np.sum(np.square(csSort2B - csSort2A), axis=1)))
        xMax2 = abs(np.min(csSort2A.T[0]))

# Filters out bad data by eliminating excessive ratios
# Filters out outliers by calculating circumference from smaller half
# The scanning process is more likely to distort a model outward than inward
    if ((.75 * sCirc2)  < sCirc1 <= sCirc2) and (.9 < (xMax1 / xMax2) < 1.1):
        circ = sCirc1 * 2
    elif ((.75 * sCirc1) < sCirc2 <= sCirc1) and (.9 < (xMax2 / xMax1) < 1.1):
        circ = sCirc2 * 2
    else:
        circ = 0

# Returns the best approximation of a the circumference of a cross section
# Returns 0 if the input data is determined to be invalid or low quality
    return circ
```

The circStack function uses the circumference function to calculate a series of horizontal circumferences in .002 (2mm) increments within a specified vertical range of its 3D input data. These circumferences are stored and returned in a list for further processing.

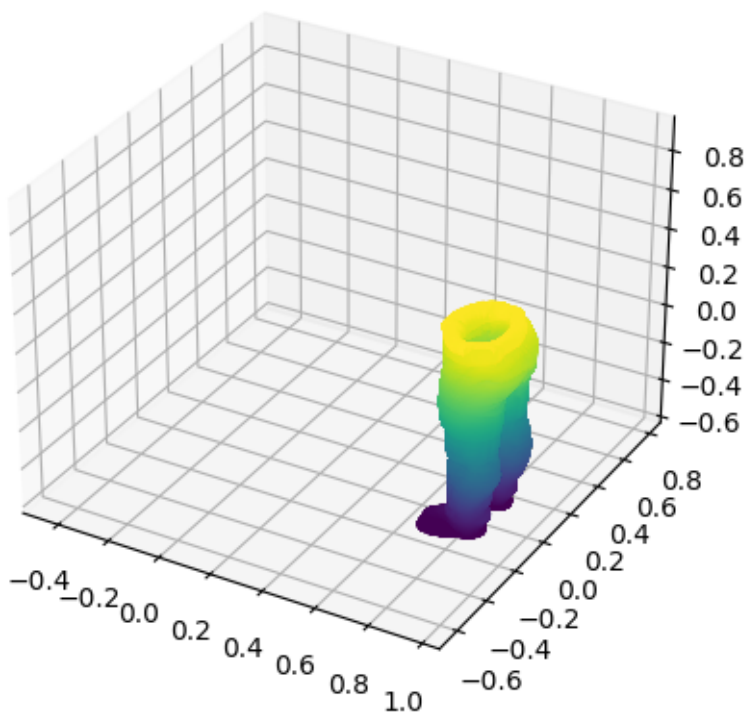
[6]:
```
def circStack(top, bot, cloud):

# Calculates number of times to call circumference(), initializes variables
    stepCount = int((top - bot) / .002)
    circs = [0]

# Iterates through input data to calculate and store circumferences at each␣
 ↪height
    for step in range(stepCount):
        crossSectionHeight = top - (step * .002)
        csPts = cloud[cloud[:, 2] == crossSectionHeight, :2]
        circs.append(circumference(csPts))

# Returns list of circumferences (floating point values)
    return circs
```

The visualize function renders a 3D scatter plot of the input data in the orientation specified by the three axis arguments. This is used to manually test, observe, understand, and prepare the initial point cloud for processing. This function is more complex than creating a 3D graph figure normally would be, because it must optimize the visual position and size of the 3D model data without distorting its proportions.

[7]:
```
def visualize(data, ax1, ax2, ax3):

    axMax = data.max()
    axMin = data.min()

    centerX = data[0].max() - abs(data[0].min())
    centerY = data[1].max() - abs(data[1].min())
    halfSpan = (abs(axMax) + abs(axMin))/2

    x = data[ax1]
    y = data[ax2]
    z = data[ax3]

    fig = plt.figure(figsize=(5, 5), dpi=100)
    ax = fig.add_subplot(111, projection='3d')
    ax.scatter(x, y, z, c=z, alpha=1)

    ax.set_xlim([centerX - halfSpan, centerX + halfSpan])
    ax.set_ylim([centerY - halfSpan, centerY + halfSpan])
    ax.set_zlim([axMin, axMax])

    plt.show()
```

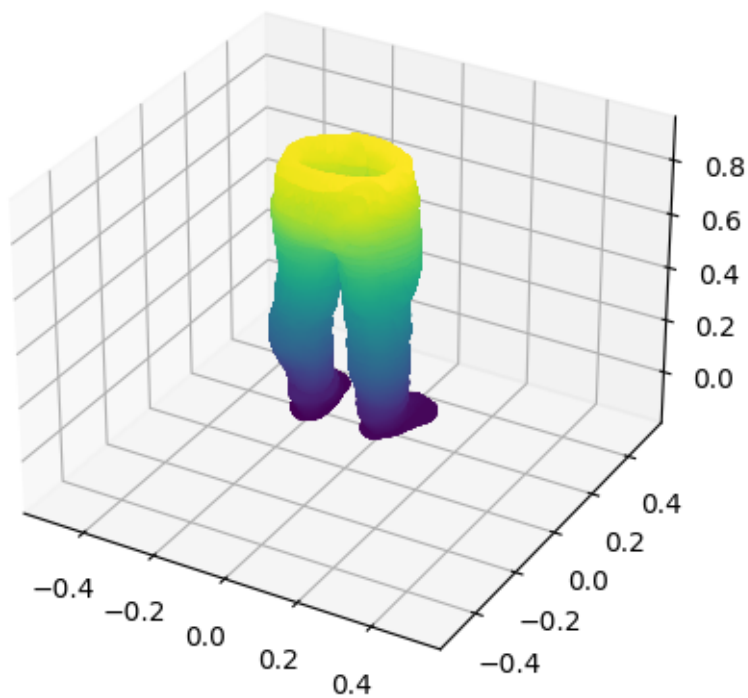
The following section begins the data mining pipelineby importing A 3D image from a file. The image was captured with the PolyCam photogrammetry app on a Pixel 7 smartphone with Android 13. The image was also cropped and roughly oriented with simple editing tools integrated into the app, then exported to one of many available file formats. As the .ply file format contains a great deal of additional data, Open3D was used to extract only the 3D coordinates of each point/vertex.

[8]:
```
# Imports input point cloud from a file using open3D
cloud = o3d.io.read_point_cloud('pants5.ply')

# Converts data format for processing
npCloud = np.asarray(cloud.points).T

# Initial visualization of raw data
visualize(npCloud, 0, 1, 2)
```



Principal component analysis (PCA) is used here to center the input data around the origin (0, 0, 0) and align it with each axis. The direction of greatest variance is calculated, then the next perpendicular direction of greatest variance, then the next. The data is then transformed such that those directions align with a set of axes. In this case, external tools were able to orient the data relatively well. However, for this purpose of detailed measurement, it was necessary for the positioning and orientation of the input data to be exhaustively precise.

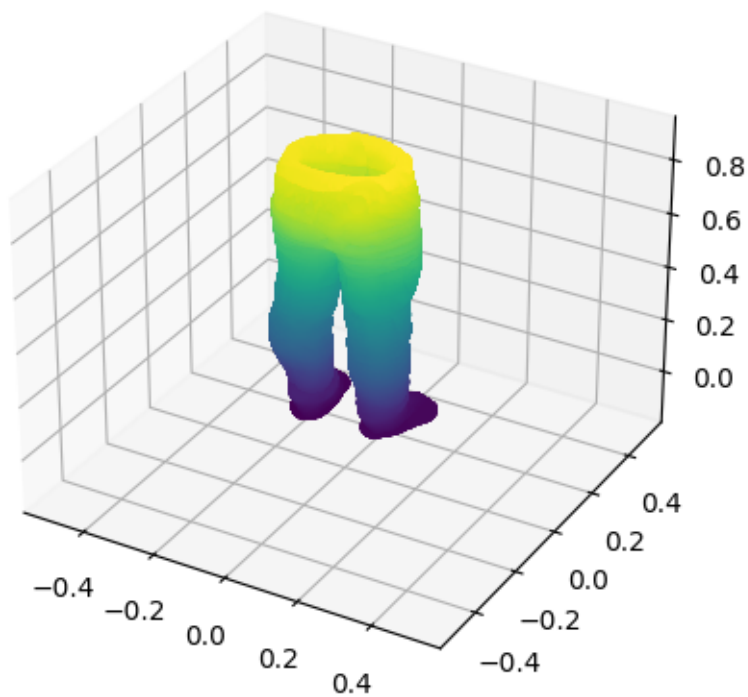
[9]:
```
# Initialized to operate on three axes/components: X, Y, Z
pca3tool = PCA(n_components=3)

# Transform data using PCA, then reorganize columns/axes to preferred upright␣
↪orientation
npCloud2 = pca3tool.fit_transform(npCloud.T).T
npCloud2 = np.array([npCloud2[1], npCloud2[2], np.negative(npCloud2[0])])
npCloud2[2] = np.add(abs(npCloud2[2].min()), npCloud2[2])

# Confirm success and check for next steps
visualize(npCloud2, 0, 1, 2)
checkMeasures(npCloud2)
```




```
Width is: 356 mm
Length is: 263 mm
Height is: 935 mm
```

Several measurements necessary for tailoring clothes follow the curved edge of a planar cross section of the 3D input data. However, coordinate points are stored in the raw data as floating point numbers, which innately contain many digits of effectively random noise beyond their actual precision. Therefore, to extract a planar cross section of the data by isolating points with certain identical values, it is necessary to truncate all of the input data down to realistic precision. The input data was produced by a 3D imaging method with a resolution of approximately one millimeter, and micrometer precision should not be necessary for clothing. The following truncates the input data (in meters) to three decimal places (millimeters).

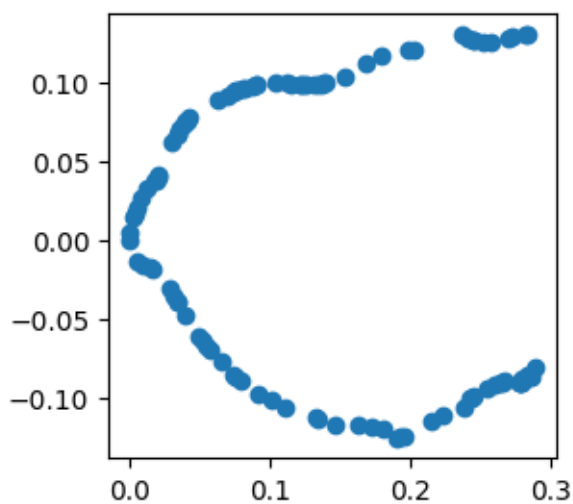
[10]:
```
# Truncates data such that points less than 1mm apart are in the same plane for␣
↪calculations
npCloud3 = (np.trunc(npCloud2*10000)/10000).T
outseam = np.max(npCloud3)

# Confirms minimal or insignificant distortion of data due to truncation
visualize(npCloud3.T, 0, 1, 2)

# Demonstrates difference before and after truncation
print("Height value before truncation:", np.max(npCloud2))
print("Height value after truncation:", np.max(npCloud3))
```




```
Height value before truncation: 0.9350571921041264
Height value after truncation: 0.935
```

The stride of a pair of pants is measured from the waistline down, between the legs, and back up to the waistline. This measurement is necessary to ensure that there is enough space below the waistline, but not so much that walking or other motions strain or tear the seam it is measured along. In this pipeline, the lowest point of the (processed) stride cross section also represents the inseam length, which is measured from the ground. The following section isolates the stride cross section for visual inspection and preprocessing.

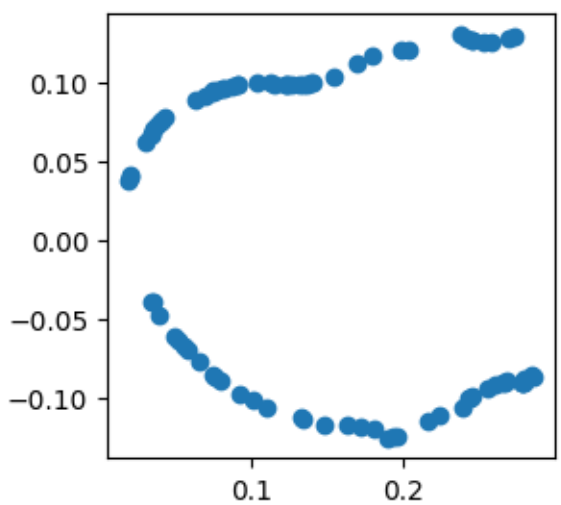
[11]:
```
# Takes cross section of point cloud necessary to measure stride
stridePoints = npCloud3[npCloud3[:, 0] == 0, 1:3]

# Checks approximate inseam length before processing
inseamRough = np.min(stridePoints.T[1])

# Pre-processes cross section by orienting and positioning it
stridePoints.T[1] = stridePoints.T[1] - np.min(stridePoints.T[1])
stridePoints = np.flip(stridePoints, axis=1)
flatVisualize(stridePoints.T)

# Prints rough inseam value for reference
print("Rough inseam value:", inseamRough)
```




```
Rough inseam value: 0.6388
```

The following section refines, splits, sorts, recombines, and prepares the stride cross section for measurement calculation.

[12]:
```
# Separates and sorts points with positive Y values from cross section
# Filters out points near Y=0 because the 3D imaging method is likely to␣
↪distort surface data outward between the thighs
spPos = np.flip(stridePoints[stridePoints[:, 1] > .03, :2], axis=0)
spPosSort = spPos[spPos[:, 0].argsort()][2 : -2]

# Separates and sorts points with negative Y values from cross section
# Filters points near Y=0 because the 3D imaging method is likely to distort␣
↪surface data outward between the thighs
spNeg = np.flip(stridePoints[stridePoints[:, 1] < -.03, :2], axis=0)
spNegSort = np.flip(spNeg[spNeg[:, 0].argsort()], axis=0)[2 : -2]

# Re-combines stride points sorted in clockwise order
# Rolls combined array to align each point in spSort with the next in spSort2
spSort = np.append(spNegSort, spPosSort, axis=0)
spSort2 = np.roll(spSort, 1, axis=0)

# Shows difference in data after processing and filtering
flatVisualize(spSort2.T)
```



The following section calculates final measurements for both the stride and inseam. Matrix operations are used to simultaneously calculate the euclidean distance between every sorted point in the cross section and the next.

[13]:
```
# Calculates more reliable inseam length value
inseam = inseamRough + np.min(spSort.T[0])
print("Inseam:", inseam)

# Calculates euclidean distance between all points
stridePartial = np.square(spSort2 - spSort).T
stridePartial2 = np.sqrt(stridePartial[0] + stridePartial[1])

# Calculates the total distance between all points but the first and last
stride = np.trunc(sum(stridePartial2[1 : ]) * 10000) / 10000
print("Stride:", stride)
```

```
Inseam: 0.6578
Stride: 0.6621
```

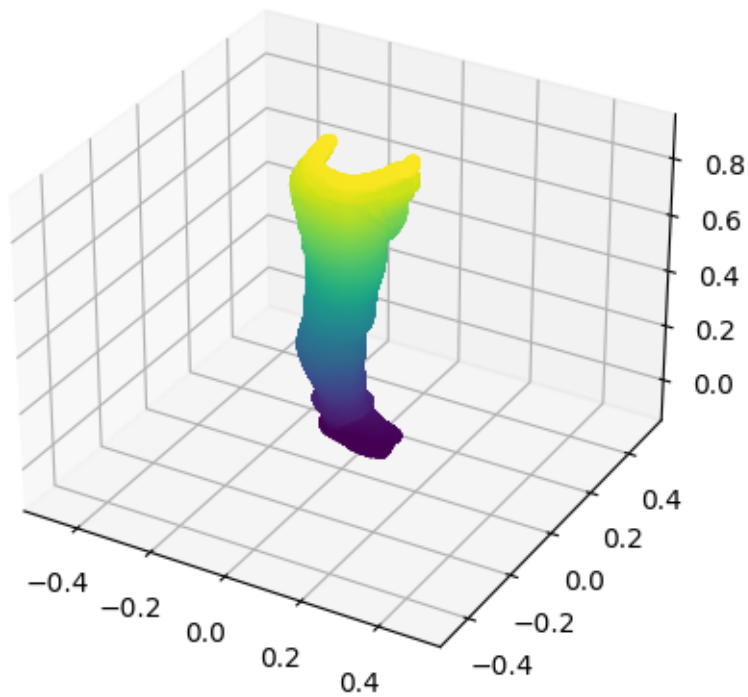
The following section searches for and calculates the waist and hip measurements. The smallest circumference very close to the top of the cropped 3D image should represent the waist. Conversely, the greatest circumference between the top of the image and the height of the inseam should represent the hip measurement.

[14]:
```
# Calculates minimum and maximum heights for hip and waist measurements
# Collects all circumferences within possible range, above the legs
whCircs = circStack(outseam, outseam - (((outseam - inseam) * .525)), npCloud3)

# Extracts the top 3mm of circumferences for the waist value
# Filters circumference lists for valid, non-zero elements
whCircsTop = whCircs[0:15]
whCircsTop = [c for c in whCircsTop if c != 0]
whCircs    = [c for c in whCircs if c != 0]

# Calculates the waistline as the minimum valid circumference in the top 3mm
# Calculates the hip as the maximum valid circumference above the legs
waist = np.trunc(min(whCircsTop) * 10000) / 10000
hip = np.trunc(max(whCircs) * 10000) / 10000

print("Waist:", waist)
print("Hip:", hip)
```

```
Waist: 0.7441
Hip: 0.9545
```

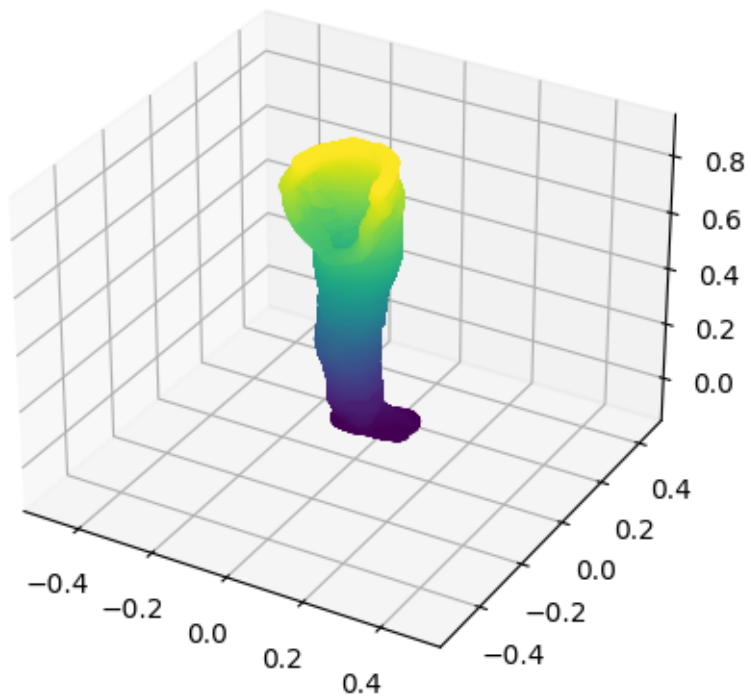
The following section isolates, positions, and orients the points of the left leg for further processing. This consists of roughly the same operations which were initially used to preprocess the full dataset.

[15]:
```
# Processes data to properly position and orient the left leg for processing
npCloud4A    = npCloud2.T[npCloud2.T[:, 0] < 0, :3]
npCloud4A    = pca3tool.fit_transform(npCloud4A).T
npCloud5A    = np.array([npCloud4A[1], npCloud4A[2], np.negative(npCloud4A[0])])
npCloud5A[2] = np.add(abs(npCloud5A[2].min()), npCloud5A[2])
npCloud5A    = (np.trunc(npCloud5A * 10000) / 10000).T

visualize(npCloud5A.T, 0, 1, 2)
```



The following section repeats the previous sequence of steps to preprocess the points of the right leg.

[16]:
```
# Processes data to properly position and orient the right leg for processing
npCloud4B    = npCloud2.T[npCloud2.T[:, 0] > 0, :3]
npCloud4B    = pca3tool.fit_transform(npCloud4B).T
npCloud5B    = np.array([npCloud4B[1], np.negative(npCloud4B[2]), np.
 ↪negative(npCloud4B[0])])
npCloud5B[2] = np.add(abs(npCloud5B[2].min()), npCloud5B[2])
npCloud5B    = (np.trunc(npCloud5B * 10000) / 10000).T

visualize(npCloud5B.T, 0, 1, 2)
```



The following section searches for and calculates the thigh measurement. A series of circumferences are calculated separately for each leg within a vertical. A relatively wide vertical range is calculated from rough human body proportions in order to correctly measure diverse body shapes. The largest circumference between both thigh regions is kept as the target measurement.

[17]:
```
# Uses approximate general human proportions to calculate a wide vertical range␣
↪where the widest part of the thigh might be
thTop = np.trunc((inseam * .95 ) * 10000) / 10000
thBot = np.trunc((inseam * .75 ) * 10000) / 10000
stepCount = int((thTop - thBot) / .002)

# Calculates the largest circumference within the above range for each leg
# Keeps the higher value as likely more accurate and less risky
thighA = max([c for c in circStack(thTop, thBot, npCloud5A) if c != 0])
thighB = max([c for c in circStack(thTop, thBot, npCloud5B) if c != 0])
thigh = np.trunc(max(thighA, thighB) * 10000) / 10000

print("Minimum & maximum location of measurement:", thBot, "to", thTop)
print("Thigh:", thigh)
```

```
Minimum & maximum location of measurement: 0.4933 to 0.6249
Thigh: 0.4838
```

The following section searches for and calculates the knee measurement. As above, a series of circumferences are calculated for each leg. The knee should be at or near the smallest cross section of the leg, and located within a narrow range. Therefore, the minimum circumference within that region of each leg is found, then the larger is kept as likely a better representation of the target measurement.

[18]:
```
# Uses approximate general human proportions to calculate a probable vertical␣
↪range where the knee might be
knTop = np.trunc((inseam * .77 ) * 10000) / 10000
knBot = np.trunc((inseam * .73 ) * 10000) / 10000
stepCount = int((knTop - knBot) / .002)

# Calculates the smallest circumference within the above range for each leg
# Keeps the higher value as likely more accurate and less risky
kneeA = min([c for c in circStack(knTop, knBot, npCloud5A) if c != 0])
kneeB = min([c for c in circStack(knTop, knBot, npCloud5B) if c != 0])
knee = np.trunc(max(kneeA, kneeB) * 10000) / 10000

print("Minimum & maximum vertical location of measurement:", knBot, "to", knTop)
print("Knee:", knee)
```

```
Minimum & maximum vertical location of measurement: 0.4801 to 0.5065
Knee: 0.3523
```

The following section searches for and calculates the shin measurement. Again, circumferences are calculated along each leg in a target region. The largest circumference between both shin regions is kept as the target measurement.

[19]:
```
# Uses approximate general human proportions to calculate a wide vertical range␣
↪where the shin might be
shTop = np.trunc((inseam * .70 ) * 10000) / 10000
shBot = np.trunc((inseam * .20 ) * 10000) / 10000
stepCount = int((shTop - shBot) / .002)

# Calculates the largest circumference within the above range for each leg
# Keeps the higher value as likely more accurate and less risky
shinA = max([c for c in circStack(shTop, shBot, npCloud5A) if c != 0])
shinB = max([c for c in circStack(shTop, shBot, npCloud5B) if c != 0])
shin = np.trunc(max(shinA, shinB) * 10000) / 10000

print("Minimum & maximum vertical location of measurement:", shBot, "to", shTop)
print("Shin:", shin)
```

```
Minimum & maximum vertical location of measurement: 0.1315 to 0.4604
Shin: 0.3631
```

The following section searches for and calculates the ankle measurement. Another series of circumferences is calculated for each leg. The ankle should be slightly above the ground but below the shin region, and at or near the wider joint itself. Additionally, it is difficult for a person to thoroughly capture this distant region of their body with the tested 3D imaging method, which may introduce inward distortion to the data. Therefore, the maximum circumferences in this region of each leg are found, but the smaller one is kept to better represent the target measurement.

[20]:
```
# Uses approximate general human proportions to calculate a probable vertical␣
↪range where the ankle might be
akTop = np.trunc((inseam * .20 ) * 10000) / 10000
akBot = np.trunc((inseam * .05 ) * 10000) / 10000
stepCount = int((akTop - akBot) / .002)

# Calculates the largest circumference within the above range for each leg
# Keeps the smaller value as likely more accurate due to probable distortion in␣
↪this area from the 3D imaging process
ankleA = max([c for c in circStack(akTop, akBot, npCloud5A) if c != 0])
ankleB = max([c for c in circStack(akTop, akBot, npCloud5B) if c != 0])
ankle = np.trunc(min(ankleA, ankleB) * 10000) / 10000

print("Minimum & maximum vertical location of measurement:", akBot, "to", akTop)
print("Ankle:", ankle)
```

```
Minimum & maximum vertical location of measurement: 0.0328 to 0.1315
Ankle: 0.2255
```

The following section collects all calculated measurements, as well as the regions in which certain measurements are applicable, and adjusts how the measurements are printed.

[21]:
```
# Grouping measurements for final processing
truncList = np.array([waist, hip, inseam, outseam, stride, thigh,
                      thTop, thBot, knee, knTop, knBot, shin, shTop,
                      shBot, ankle, akTop, akBot])

# Truncating all measurements for presentation
tr = np.trunc(truncList * 10000) / 100
```

The following section outputs all calculated measurements in an organized format.

[22]:
```
# Final output

print("All measurements are in centimeters")
print(" ")

print("Waist        ", tr[0] )
print("Hip          ", tr[1] )
print("Inseam       ", tr[2] )
print("Outseam      ", tr[3] )
print("Stride       ", tr[4] )
print(" ")


print("Measurements are located between the stated top and bottom heights")
print(" ")

print("Thigh        ", tr[5] )
print("Thigh top    ", tr[6] )
print("Thigh bottom ", tr[7] )
print(" ")

print("Knee         ", tr[8] )
print("Knee top     ", tr[9] )
print("Knee bottom  ", tr[10])
print(" ")

print("Shin         ", tr[11])
print("Shin top     ", tr[12])
print("Shin bottom  ", tr[13])
print(" ")

print("Ankle        ", tr[14])
print("Ankle top    ", tr[15])
print("Ankle bottom ", tr[16])
```

All measurements are in centimeters

Waist	74.41
Hip	95.45
Inseam	65.78
Outseam	93.5
Stride	66.21

Measurements are located between the stated top and bottom heights

Thigh	48.38
Thigh top	62.49
Thigh bottom	49.33
Knee	35.23
Knee top	50.64
Knee bottom	48.01
Shin	36.31
Shin top	46.04
Shin bottom	13.15
Ankle	22.55
Ankle top	13.15
Ankle bottom	3.28

END – Software Deliverable

BEGIN – Prototype Product Deliverable

PROTOTYPE PRODUCT

Specifications

Product

The prototype product deliverable is one fitted pair of pants. This garment was ordered from a seamstress with measurements exclusively produced by the software deliverable. It is to be physically presented for inspection.

Purpose

The primary purpose of the prototype product is to validate the other two deliverables. A business model relies on a feasible product, and the software deliverable must be complete to create that product.

As it was not a purpose of this deliverable to demonstrate the acquisition or mastery of sewing skills, assembly of the prototype was outsourced to a seamstress. This allowed the prototype to fulfill the additional purpose of simulating the proposed flexible manufacturing process for similar products.

In general, the prototype product should demonstrate that it is feasible to automate the tailoring process using existing 3D imaging software and flexible manufacturing technology.

Conditions

The product is constructed in strict adherence to the measurements produced by the software deliverable and a predefined standard template, with no quantitative human input.

Qualitative choices are specified in a simple format representative of multiple choice selections in an online ordering process. The type of standard template, material, color, texture, basic features, and type of fit were presented as part of the order. However, all qualitative choices were marked as flexible, subject to the priority of on-schedule order fulfillment.

The product is produced by a professional seamstress with only the information which would be given to a flexible manufacturing machine, in order to simulate the process.

The prototype product demonstrates proof of concept that a business with commercial resources would likely be able to produce a fully consumer-ready product.

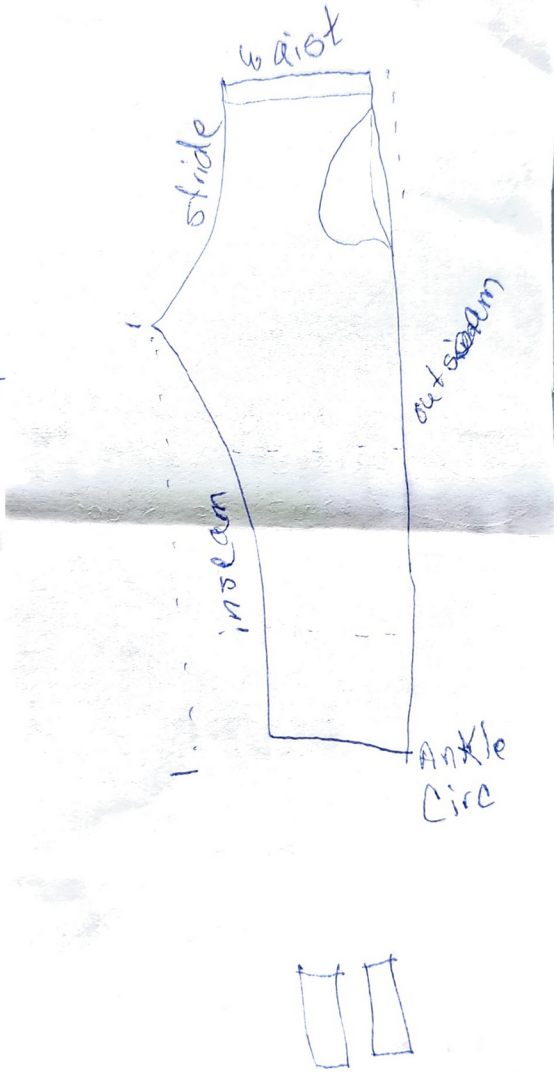
Measurements

A local small business (Sewing Shop Bozeman 2023) was contacted to ascertain a list of measurements necessary to fully tailor a pair of pants and how to collect them. Those measurements are the waist, hip, inseam, outseam, and ankle, then optionally the height, weight, and/or additional circumferences along the leg. Finally, ease was described as a positive or negative value to adjust the other values for tightness or looseness, with certain default values for different types of garment or fabric. The following document and notes, front and back, was provided after a consultation:

The Sewing Shop Bozeman Intake Form

Date	**Time**
Name	
Height	
Weight	
Neck	
Arm-Bent	
Arm-Straight	
Chest/Bust	
Waist	where you want waist band
Hip	widest part
Outseam	waist to floor
Inseam	crotch to floor
Head	
Around shoulders	
Notch to floor	
Point-to-Point	
Over Bust	
Under Bust	
Stride - waist under legs to waist	
Ankle circum	

Ease →



RESULTS – Prototype Product Deliverable

The prototype product deliverable must be presented separately from this document, as it is a physical object. Through a local small business (Sewing Shop Bozeman 2023), a seamstress was hired to construct a pair of pants from only the measurements produced by the software deliverable.

During the final stage of development of the software deliverable, inquiries about placing a unique order were sent on March 21st to several local tailoring and sewing businesses. The goals and conditions of this project were described in each inquiry. The only response was received the same day, and a consultation appointment was scheduled for March 29th to discuss the project and order in greater detail.

The project was received well during the consultation, and the seamstress generously explained in detail exactly which measurements are necessary for tailoring, why, and how to take them. A timeline for the order was established, with a deadline of April 6th to provide all necessary measurements. The software deliverable was successfully completed on April 5th, and its exact printed output of the measurements was sent along with the following preferences:

Rough design:	Slacks
Material:	Woven
Color:	Textured grey
Texture:	Smooth / soft
Belt loops:	Yes
Pockets:	Sides, deep
Ease:	Template standard for low but not tight

These additional specifications were limited to simple qualitative options which could be collected using a simple digital multiple choice form. In order to simulate an automatic flexible manufacturing process, no other information was provided or collected to complete the order.

A call was placed to confirm the order, as well as receipt of the specifications necessary to fulfill it. An order completion date of May 3rd or earlier was established in order to present the prototype product deliverable on May 5th. A second call on April 26th confirmed that the order was on schedule to be completed and picked up before May 5th.

END – Prototype Product Deliverable

BEGIN – Business Model Deliverable

BUSINESS MODEL

Specifications

Purpose

The purpose of the business model deliverable is to demonstrate, by any appropriate means, the design, considerations, and prospects of a business which could extend and apply the software and product deliverables to a commercial scale.

Terminology

This deliverable was described in the original proposal as a 'business plan' with the intent of referring to a general category of methods and documents. However, it was clarified later in the project that the phrase 'business plan' refers to a specific type of document and prescriptive method of estimation within that category. Instead, a type of 'business model' was selected for this deliverable as the most effective vehicle of its intended purpose.

As the approved deliverables were the most permanent and inviolable aspects of this project, it is important to clarify this change in terminology. The purpose statement above reflects the unchanged, original intended purpose of this deliverable.

Format

Traditionally, a business plan is created before an entrepreneurial venture is established or tested. The prospective business owner exhaustively defines all aspects of the venture, and personally projects the first several years of its finances.

A business plan includes the following sections of definitions and predictions: executive summary, company description, service or product line, market analysis, marketing and sales, financial projections, organization and management, funding request, and appendix (United States Small Business Administration n.d.). Although a business plan is composed from a variety of documents which are useful for the operation of established businesses, it is quite vulnerable to cascading unforeseen circumstances.

By contrast, the Lean Startup methodology iteratively develops and tests a business model with extensive feedback from potential customers. The result is a brief but highly refined business model canvas which represents high level concepts and tested hypotheses. This format was selected because it is optimized to test and refine the concept and viability of an unrealized business, rather than roughly forecasting the operational details of a concretely defined business. Additionally, an opportunity was presented in another project to simulate the Lean Startup process, and thereby jump-start the development of the business model deliverable.

Structure

A business model canvas is composed as a visual summary with nine categories of interconnected design choices: value proposition, customer segments, customer relationships, channels, revenue streams, cost structure, key resources, key activities, and key partners. The canvas is iteratively developed by a new company, after its inception, as it searches for a viable purpose and business model. The following diagram is the standard business model canvas layout:

<table>
<tr><td rowspan="2">Key Partners</td><td>Key Activities</td><td rowspan="2">Value Proposition</td><td>Customer Relationships</td><td rowspan="2">Customer Segments</td></tr>
<tr><td>Key Resources</td><td>Channels</td></tr>
<tr><td colspan="3">Cost Structure</td><td colspan="2">Revenue Streams</td></tr>
</table>

The extended business model includes a set of validated hypotheses about each design choice in the business model canvas, organized by the same categories. Without testing and validation, all claims and explanations are considered to be unreliable assumptions. The Lean Startup methodology specifies that these hypotheses can only be validated through feedback from real potential customers and business partners, with support from research and expert opinions.

Methods

Business Modeling

The business model deliverable was developed using (a simulation of) the Lean Startup methodology (Blank 2013). This process consists of the following:

Initially:
1. Develop a low fidelity business model canvas
2. Produce hypotheses about each section of the business model canvas

Repeatedly:
3. Test model hypotheses with feedback from potential customers
4. Revise, validate, or pivot to new hypotheses
5. Develop a prototype, minimum viable product
6. Test and refine the product with feedback from potential customers

Finally:
7. Consolidate, refine, and finalize business model canvas
8. Transition from ‘start-up mode’ to an established, traditional business model

The Lean Startup methodology is based on Agile software development and Lean Manufacturing. Instead of using or working toward a traditional business plan, it focuses on dynamically and iteratively developing a business model using customer feedback. It emphasizes that any plan or model developed by an entrepreneur in isolation is merely a set of untested hypotheses. To test and refine those hypotheses, the Lean Startup methodology uses the customer development process. The entrepreneur or company searches for potential customers of their current business model, interacts with them to test hypotheses and prototype products, then refines the business model with that feedback.

Simulation

An opportunity was presented and leveraged to develop the business model deliverable during an eight week course project. The business concept from this project was proposed alongside others, using the two following documents, as a hypothetical startup company named "MeasureTech". This startup was selected by popularity for further development, then operated and tested in a close approximation of the Lean Startup process for a real company. Six one-week iterations of the customer development process were conducted, where each iteration focused on hypotheses about different parts of the business model canvas.

The following sections detail the startup proposal, and then weekly iterations of the MeasureTech concept which was used to develop the business model deliverable.

Concept Development History

Initial Idea

This concept was initially created for the MSU Bozeman Directed Interdisciplinary Studies program as a replacement for another capstone project idea. The original idea was for a monolithic solution to one large abstract problem. The replacement was inspired to be the opposite of that in every way, by using new but existing technology to flexibly solve small but ever-present inconveniences in life for a wide variety of people. This ideal was refined into one project which aimed to solve a problem for many different people of finding affordable clothing in their size and shape.

Version 1

The very first version of this idea was a simple thought that a problem might have a solution. This idea was considered for a few weeks to identify possible solutions. At this stage, the idea had minimal impact and zero feasibility.

Version 2

The second version of this idea included a possible, partial solution: using depth cameras, 3D modeling software, and some kind of 3D-to-2D software to fit clothing to people. This version generalized the idea from one person to a solution for many different people, which in turn may be viable for a business to implement. Research indicated that this solution was theoretically feasible, and the concept was received with positivity, interest, questions, and ideas. At this stage, the idea had low impact and low feasibility.

Version 3

The concept was refined, adjusted to fit D.I.S. requirements, proposed for approval, and finalized in its third version as a D.I.S. capstone project. A curriculum was designed, proposed, and approved to support the project. At this stage, the idea had low impact and moderate feasibility.

Version 4

In order to continue work on this concept, a variety of interdisciplinary skills were developed over two years. Although the general concept and final goals were set in stone with the previous version, research and adjustments for other aspects of the idea continued. People with sensory sensitivities were identified as an important group for the concept to support. The fourth version of the idea included a few details of a business model, a distinct target market, and potential technologies to use. At this stage, the idea had moderate impact and moderate feasibility.

Version 5

Three concurrent courses dramatically reshaped the fifth version of this concept. With knowledge of small business management, the idea was refined to describe a business startup with a target market, sales models, and potential product lines. With knowledge of data mining, the technical requirements shifted from advanced 3D modeling software toward relatively simple data mining techniques. Additionally, the originally proposed 3D imaging hardware and software were tested, and found to be insufficient. The fifth version of the idea incorporated these new details and future directions. At this stage, the idea had good impact and moderate feasibility.

Version 6

The concept was reviewed under new oversight, slightly redefined, and reaffirmed with its new methods, technology, and final expectations. The sixth version of the idea contextualizes it within the tailoring services industry, then integrates and summarizes the changes to the idea since version 3. At this stage, the idea had good impact and good feasibility.

Version 7

For this proposal, the concept was reformatted as a hypothetical business startup, by resolving a number of design decisions and establishing more comprehensive and specific details throughout. Version 7 of the idea is the following Startup Concept Proposal document. At this stage, the idea was estimated to achieve relatively high impact and feasibility.

Startup Concept

Idea Pitch – MeasureTech

Only a small fraction of manufactured clothing fits any one individual well, even in what should be the right size, and existing clothes-tailoring services are expensive due to the manual labor and skill required. This startup will use 3D imaging technology to automate the measuring and fitting process, and use flexible manufacturing to automate the production of affordable, sensory-friendly individually tailored clothing.

Customer & Need

The ideal customer of this startup is a slightly fashion-oriented, educated young adult with sensory sensitivities and unique body proportions who lives in a city, earns a moderate income, and enjoys casual shopping, technology, and novelty. More broadly, the customers of this startup could include any person who has trouble finding clothes which fit satisfactorily well. These customers, ideal or not, have an unfulfilled need of affordable, comfortable clothing, which is rarely or never satisfied by clothes manufactured in predetermined sizes. Although specialty brands and tailoring services exist to fill a similar need for comfortable, specially sized/shaped clothing, they are not affordable compared to mass-manufactured clothing.

Unfortunately, it has not been feasible to conduct research to definitively confirm that this is a driving need in the market. However, this business concept has been frequently discussed with a wide variety of people over three years, and many have commented that they cannot find clothes which fit just right. Throughout these discussions, no skepticism, doubt, or disbelief was voiced about the existence of the unfulfilled need.

Product Offering

This startup will solve the problem of providing affordable fitted clothing by automating the tailoring process to reduce production costs. The ideal name, vision, and brand image of the business are uncertain. Hypothetically, the brand should focus on practicality, comfort, affordability, and catering to sensory sensitivities.

Sensory sensitivities may occur due to mental or physical illnesses/conditions; some examples include being unable to tolerate certain fabrics, obtrusive seams, or tight spots. Practicality and comfort should be emphasized because fashion is not currently a primary goal of the business, and affordability should be emphasized because it is the primary advantage over traditional tailoring services.

Value Proposition

The unique value which this business offers to customers is affordability. Tailoring services already exist, but the time-consuming, high-skill manual labor they require means that they can only cater to higher-income consumers. By automating the process, this startup aims to compete with mass-manufactured clothing by offering comfort and personalization at a marginal extra cost. In one brief tagline and value statement, this business offers an affordable, perfect fit.

Competitive Advantage

The biggest competitive advantage of this startup is automation. Currently, there are only a couple small competitors in its exact market niche. Therefore, it will mostly compete against products from traditional tailors and large clothing manufacturers. This startup will produce fitted clothing more affordable than products from tailors, and more comfortable than products from traditional mass manufacturers.

The business' primary distinctive competency is in 3D imaging technology and measurement. This competency is important because it may be used to branch or pivot into other market niches. As well as clothing, the same measurement technology could be applied to fitting footwear, sports gear, protective equipment, prosthetics, or computer peripherals. Additionally, it could be used to offer measurement-matching services to both consumers and businesses in order to improve online shopping experiences. Distinctive competency in 3D imaging technology and measurement is currently almost exclusively held by business-to-business companies operating in the area of high budget precision machining. The possible startup name "MeasureTech" refers to this competency, and could encompass a business structure with several focused subsidiary companies.

Opportunity Analysis

This meets the threshold for a good new startup opportunity because has consistently received level-headed positive feedback from professors, advisors, friends, family, and acquaintances for three years. There have not been any negative or disinterested responses, nor any alarmingly overzealous or excessively excited responses. Instead, the responses have been a consistent blend of intrigue, questions, and collaborative discussion of opportunities and challenges.

Revenue model

There are two primary ways this product may be offered. Using smartphone-based 3D imaging software, the business will be able to operate entirely online. 3D images would be taken by customers, processed into measurements, then used to produce clothing to be shipped to them. Alternatively or additionally, the business may operate physical stores or booths where customers can have their measurements taken, order clothing, and then have it shipped to them. The current price goal for products is 10-25% above non-fitted clothing due to value added, where traditional tailoring may cost 100-200% more due to manual labor costs. Feedback to date on the concept indicates that this revenue model should be feasible and profitable.

Key Concerns

3D imaging technology may exhibit unusual limitations when used for taking measurements. For example, a physical store location may require dressing rooms, laundry machines, and form-fitting patterned clothing for customers to wear while taking measurements. Additionally, this new application of relatively new technology poses a risk that people may not believe a low-skill, automatic system could take precise measurements to tailor clothing.

Initial Hypotheses

From the startup concept proposal, a set of untested hypotheses was drafted for each category of a business model canvas. It was understood that these hypotheses were not comprehensive, and that any number of them could be entirely incorrect. The following hypotheses were all produced before iterative testing began:

Value Proposition
- Products should be affordable for many customers
- Comfort should be prioritized
- Products should accommodate unique needs
- Customers would value transparent material sourcing

Customer Segments
- People who want well-fitted clothing
- People with unique sensory or other needs for how clothing fits
- People who do not want to search for affordable clothing which fits them

Customer Relationships
- Customer relationships would be close and extended
- Customers would return for reliably fitted products

Revenue Streams
- The primary revenue stream would be clothing sales to individual customers
- Prices can be higher than mass-manufactured clothing due to added value
- Prices can be low enough to compete with mass manufactured products
- Licensing out 3D imaging and processing software could also provide revenue
- Measurement services for online clothing retailers could also provide revenue

Cost Structure
- Fabric, and other clothing materials would be the primary variable costs
- Research and development would be a fixed cost
- Manufacturing equipment would be a fixed cost
- Rent could be another necessary cost

Channels
- The first sales channel should be online to reduce initial costs
- Customers would be willing to take their measurements using a phone camera
- Customers would order online with a file and set of qualitative preferences
- A store location could be another viable channel
- Mall booths could be another viable channel

Key Activities

- Finding new applications of 3D imaging software could drive business growth
- It would be important to continue refining the 3D imaging technology in use

Key Resources

- 3D imaging tools would be acceptable to potential customers
- Customers would prefer and adopt smartphone-based 3D imaging tools
- Flexible manufacturing technology would be necessary to produce clothing
- Manufacturing materials would be important to maintain supply of
- 3D data processing software would be required as well as 3D imaging tools

Key Partners

- A social media marketing firm could reach the most potential customers
- Partnerships with other clothing brands and businesses would be beneficial
- 3D imaging technology businesses could be important partners

Feedback on these hypotheses was collected informally and transparently as part of a course project, from a variety of people acting as potential customers of the hypothesized business. Respondents included friends, family, roommates, coworkers, online acquaintances, members of online communities, and business representatives. No obligation, pressure, or incentives were given for respondents to begin or continue to respond. No identifying information was collected or retained, and no demographic information was required. Feedback was collected through asynchronous online surveys, real-time text conversations, and personal interviews.

Iteration 1

The first iteration of the customer development process tested the foundational assumptions of the proposed business model with a high volume of survey responses. The first priority was to verify the that the proposal correctly identified a widespread unfulfilled need, and therefore that future iterations would not produce a solution without a problem. The survey also collected information about preferences between three possible sales channels to collect measurements, place orders, and deliver products. The following hypotheses were tested:

Value Proposition

1. Products should be affordable for many customers
2. Comfort should be prioritized
3. Products should accommodate unique needs
4. Customers would value transparent material sourcing

Customer Segments

5. People who want well-fitted clothing
6. People with unique sensory or other needs for how clothing fits
7. People who do not want to search for affordable clothing which fits them

Channels

8. The first sales channel should be online to reduce initial costs
9. Customers would be willing to take their measurements using a phone camera
10. Customers would order online with a file and set of qualitative preferences
11. A store location could be another viable channel
12. Mall booths could be another viable channel

Results

The tentatively identified unfulfilled need was validated, as nearly all respondents noted that they struggled to find properly fitting clothing. Many responses indicated that this difficulty arose from unique and varied body proportions, and one person reported that their clothing options were limited by the sensation of certain fabrics. A majority of responses stated that comfort was the most important aspect of clothing.

A slight majority of respondents indicated that they had justified paying for traditional tailoring services on at least one occasion. The survey responses similarly favored a traditional store location and a fully online channel, but also indicated people would not be likely to find or enter the store. Unexpectedly, precisely zero interest was expressed in a mall booth sales channel. This data supported the following changes:

Updated Hypotheses

- Validated: 8
- Uncertain: 1-7, 9, 10
- Revised: 11
- Eliminated: 12

Discussion

Overall, this survey suggested that potential customers do have an unfulfilled need for affordable fitted clothing, but it did not provide sufficient information to evaluate how well the proposed value proposition would address that need. Similarly, there was not enough data to meaningfully refine the proposed customer segments.

By a small margin, enough feedback was collected to justify cascading changes to the statuses of three sales channel hypotheses. The mall booth sales channel was quickly eliminated due to a complete lack of support, while the other two received nearly even quantitative support. Additional qualitative commentary favored the online sales channel over the traditional store channel, which would be far more expensive due to rent and labor, and no new channels could be identified. Therefore, the online sales channel hypothesis was validated as the only economically viable sales channel with customer support. As the traditional store hypothesis remained well-supported, it was revised rather than eliminated.

Overall, survey respondents in the role of potential customers stated that affordability, comfort, and a variety of clothing options would be the the most value-adding aspects of the products and business. Fortunately, the survey responses provided significant insights into more precise, directed, and useful questions to ask in later iterations, where this iteration was limited by its reliance on open-ended, general information gathering.

Iteration 2

The second iteration of the customer development process tested another set of hypotheses with another online survey. This survey was designed to refine the general information from the first survey with concrete data, and distributed to a diverse range of potential customers with experience paying for tailoring services. The questions focused on how much potential customers have paid or would be willing to pay for a range of different fitted garments. This iteration tested the following hypotheses:

Revenue Streams

1. The primary revenue stream would be clothing sales to individual customers
2. Prices can be higher than mass-manufactured clothing due to added value
3. Prices can be low enough to compete with mass manufactured products

Value Proposition

4. Products should be affordable for many customers
5. Comfort should be prioritized
6. Products should accommodate unique needs
7. Customers would value transparent material sourcing
8. Fast tailoring and delivery would be important to some customers

Customer Segments

9. People who want well-fitted clothing due to sensory needs
10. People who want or need tailored clothing to be made quickly
11. People who do not want to search for affordable clothing which fits them
12. People who are willing but reluctant to pay the price of traditional tailoring

Results

Potential customers stated that they were most interested in tailored premium products, and that they had purchased such products before. Formal events and product appearance were the foremost reasons listed for these past purchases. The next most popular items products were work clothing, then everyday clothing. Unexpectedly, respondents were willing to pay disproportionately high margins for tailored everyday clothing such as jeans and t-shirts. Additionally, responses selected and mentioned jeans and low-cost items significantly more than other items.

These results were surprising in that each metric supported the viability of a different product. Potential customers expressed only mild interest in other products such as tailored accessories, tools, or hardware peripherals. Finally, it was noted that the survey had unfortunately not reached any members of the ideal target market. In summary, it was determined that the business model should incorporate several types of (exclusively) clothing products. All tested hypotheses were tentatively validated, pending further review during later iterations with more data.

Updated Hypotheses

- Validated: 1 – 12

Iteration 3

The third iteration of the customer development process tested another set of hypotheses with one-on-one interviews about various questions and feedback. The goal of this was to gain a better understanding of exactly what potential customers would personally think and feel about certain products. Additionally, this process was used to continuously develop better questions to ask, and to determine which hypotheses to focus on with each new iteration. This iteration tested the following hypotheses:

Channels

1. Customers would be willing to take their measurements using a phone camera
2. Customers would order online with a file and set of qualitative preferences
3. A store location could be another viable channel

Key Resources

4. 3D imaging tools would be acceptable to potential customers
5. Customers would prefer and adopt smartphone-based 3D imaging tools

Customer Relationships

6. Customer relationships would be close and extended
7. Customers would return for reliably fitted products

Results

During interviews, one potential customer emphasized the importance of fitted formal wear for looking and feeling good at work and special events, but also the value of the in-person tailoring experience. Another potential customer also stated a preference for in-person tailoring, citing concerns that invisibly captured measurements would feel less reliable. However, they also stated that a faster tailoring and delivery process would be beneficial due to time constraints and schedule conflicts. Several other potential customers, including college students, also prioritized affordability and speed over perfect-fitting clothing.

Although these results were informative, this iteration primarily identified challenges, limitations, and methods of testing hypotheses with personal interviews, from which a new hybrid system was developed to collect feedback. This system used online surveys to collect discrete data points from a broad and varied audience, and used interviews to deeply evaluate certain hypotheses and high level conceptual questions. It was determined that these interviews would be necessary to fully understand customers' problems and needs, identify customer segments with addressable needs, and confirm customers' understanding of the business and its marketing.

Updated Hypotheses

- Uncertain: 1, 2, 4-7
- Revised: 3

SWOT Assessment

After the third iteration, the business model was assessed for strengths, weaknesses, opportunities, and threats, with the following results:

- Its strengths include affordable personalized products, low production costs, and a wide range of product offerings. Additionally, its unique made-to-order manufacturing process dramatically simplifies product inventory management by eliminating excess production.

- Its weaknesses include changing fashion trends, reliance on only one or two revenue streams, and reliance on key resources which future competitors may be able to replicate. The weakness of replicable technology may be partially addressed by the key activity of continuous research and development, but the other two require further intervention.

- Its opportunities include licensing software at a premium for other clothing businesses to implement, developing an additional subscription-based revenue stream, and searching for viable non-clothing products. Leveraging these opportunities would allow the business to diversify, and therefore develop resistance to capricious market conditions or other unexpected circumstances.

- Threats to this business model include high and externally defined equipment costs, major clothing brands, other tailoring services, and other emerging custom clothing businesses using Industry 4.0 technology. The cost of flexible manufacturing equipment is an immediate concern because it strongly influences the amount of initial capital required to start the business. By contrast, the various other clothing businesses are not imminent threats because they seem to occupy different market niches from this business model.

Canvas Changes

As this iteration of feedback did not provide conclusive evidence for or against the tested hypotheses, an additional review of all prior feedback and analysis was conducted.

Upon reviewing new software development progress, software licensing was eliminated as a viable revenue stream. Additionally, independently operated physical stores were eliminated as a channel due to high costs, high labor requirements, and middling customer interest. However, a new prospect emerged for establishing a physical presence through in-store partnerships with other businesses. Hypotheses about this possibility were created for testing.

The initial hypotheses about customer relationships and key partners were slightly adjusted to be more consistent with other parts of the canvas. Finally, the customer segments and value proposition sections of the canvas were reorganized to visually align related items in order of estimated importance.

Iteration 4

The fourth iteration of the customer development process aimed to refine the previous set of hypotheses, with one adjustment, through conversations with both potential customers and members of specialty clothing businesses. Managers from Plume Bridal and Men's Wearhouse were interviewed about what they struggle with, what hazards and opportunities they have encountered, and what an Industry 4.0 business partner could do for them. Potential customers were contacted through academic and personal messaging servers to discuss the validity and implications of various hypotheses, potential directions for further development, and how to clearly present and describe the business model. This iteration re-tested the following hypotheses:

Channels

1. Customers would be willing to take their measurements using a phone camera
2. Customers would order online with a file and set of qualitative preferences
3. Store locations of partnered businesses could be another viable channel

Key Resources

4. 3D imaging tools would be acceptable to potential customers
5. Customers would prefer and adopt smartphone-based 3D imaging tools

Customer Relationships

6. Customer relationships would be close and extended
7. Customers would return for reliably fitted products

Results

The business managers identified hiring and training employees as one of their biggest challenges, and confirmed that a business using the hypothesized channels and key resources could potentially avoid that challenge. Furthermore, they expressed interest in a service or partnership to access those tools and systems, and mitigate that challenge. This introduced a clear opportunity to establish and maintain physical presences through in-store business partnerships, which supplanted the previous hypothesis about store locations.

One potential customer requested clarification about two of the channel hypotheses, stating that an app-only ordering system would be problematic for them. After further discussion, both hypotheses were validated, but an additional hypothesis was created and tested regarding support for multiple integrated online channels, such as an app and a website. The other hypotheses about online channels were well received by several people and largely validated.

Updated Hypotheses

- Validated: 1, 2, 4, 6
- Uncertain: 5, 7
- Revised: 3

Iteration 5

The fifth iteration of the customer development process began with a review of the plentiful feedback collected from extended conversations with potential customers in the previous iteration. This information contained valuable insights into how to accommodate a wide variety of unique needs and preferences including fit, fabric, seams, features, and other manual adjustments. In turn, these insights crystallized an idea from patterns in feedback since the first iteration. The loosely defined tagline of "an affordable, perfect fit" was redefined with confidence to the brand image of "total personalization, in your hands".

After this review, the fifth iteration focused on continued discussions with potential customers about hypotheses, and discussions with larger businesses such as Macy's and Nordstrom about partnerships. This iteration tested the following hypotheses:

Revenue Streams

1. Measurement services for online clothing retailers could also provide revenue
2. In-store revenue streams should be developed through business partnerships

Cost Structure

3. Fabric, and other clothing materials would be the primary variable costs
4. Research and development would be a fixed cost
5. Manufacturing equipment would be a fixed cost

Key Partners

6. A social media marketing firm could reach the most potential customers

Results

The discussions reaffirmed that both businesses and consumers are interested and viable as potential customers of the business model. Working directly with consumers would allow the business to develop very efficient revenue streams by focusing research and development on a singular system of adaptive technology. Although working with established businesses requires more software development and negotiation for different use cases, it would provide a route through robust channels to reach their existing customer base.

It was decided that working with both businesses and customers would be most beneficial. Direct purchases would provide valuable feedback and data about non-measurement choices (to maintain privacy) for refining the automated production systems. In turn, the improved technology would be leased to business partners, and a significant portion of customer relationship maintenance and marketing work would be delegated to those businesses, to mutual benefit.

Updated Hypotheses

- Validated: 1-6

Iteration 6

The sixth and final iteration of the customer development process tested all remaining untested or revised hypotheses. Some additional feedback was collected, then feedback from all iterations was organized together for a final analysis. Although many additional hypotheses were considered during the analysis, its primary purpose was to test a smaller number of already partially supported hypotheses. This iteration tested the following hypotheses:

Value Proposition
1. Products should accommodate all customers' needs and preferences

Customer Relationships
2. A focus on personalization would facilitate close customer relationships
3. Measurement and preference profiles would draw repeat customers

Key Activities
4. Finding new uses for 3D imaging software would drive business growth
5. It would be important to continue refining the 3D imaging technology in use

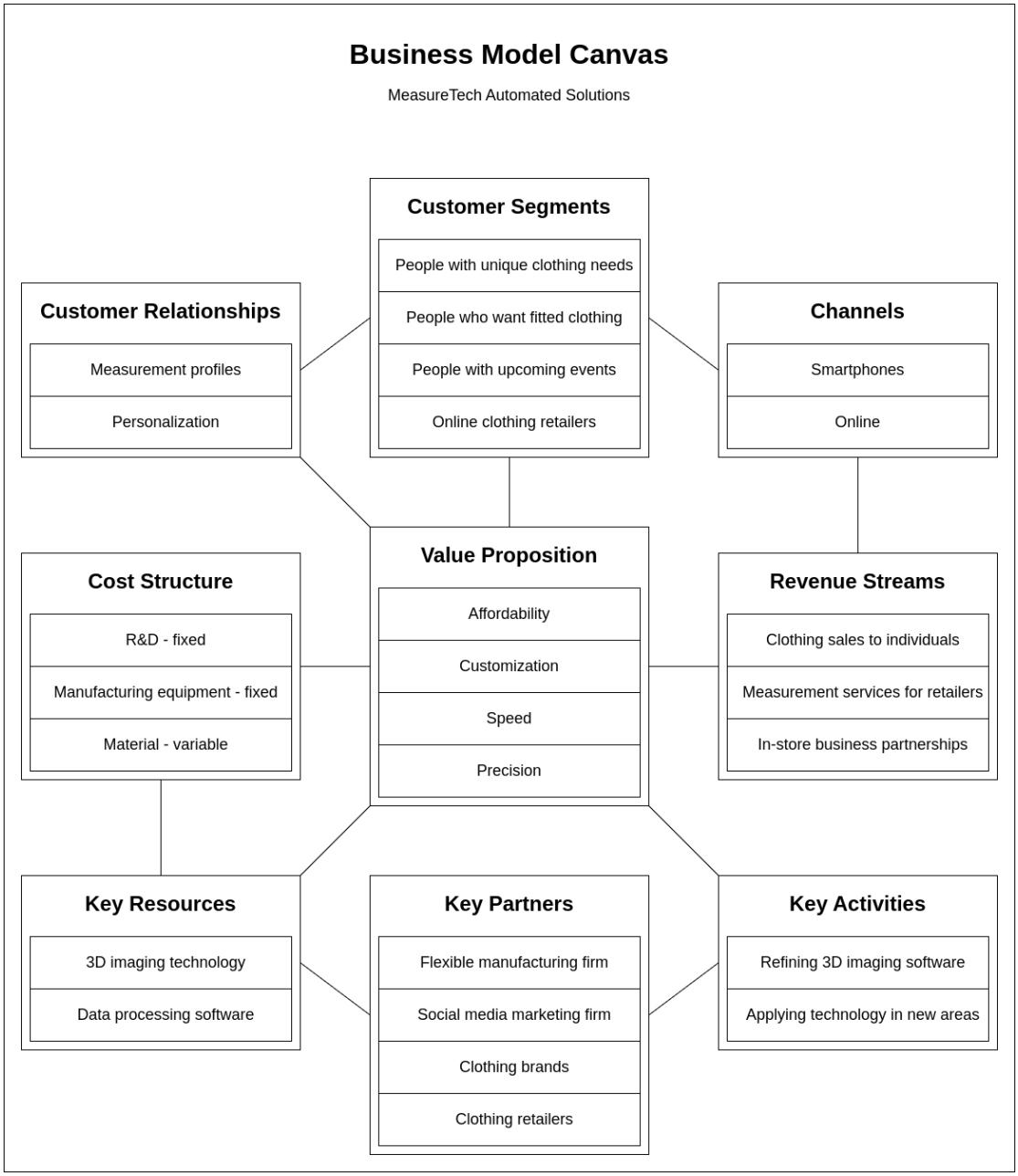
Key Partners
6. It would be necessary to work with a flexible manufacturing business
7. It would be beneficial to work with other clothing brands and businesses

Results
The closer, comprehensive analysis identified additional evidence which reinforced each of these partially supported hypotheses. As with each other iteration, the business model canvas was adjusted to reflect the newly validated hypotheses. Then, the canvas and full list of validated hypotheses supporting it were polished and finalized.

Although this business model was final, it was importantly perfected or even necessarily completed. As real startup companies eventually run out of capital and dissolve without committing to a business model, so did the simulated startup run out of time. With this iteration, the business model was estimated to be coherent, well-rounded, viable, and at minimum a viable starting point for an actual business.

Updated Hypotheses
- Validated: 1-7
- Eliminated: 8-?

RESULTS – Business Model Deliverable



This is the final business model canvas, reformatted for visual clarity on a vertical page and to visualize the unique relationships between sections of this particular business model. All contents are identical in the standard business model canvas layout below, which implies additional interactions between sections of the canvas.

<table>
<tr><td rowspan="2">Key Partners</td><td>Key Activities</td><td rowspan="2">Value Proposition</td><td>Customer Relationships</td><td rowspan="2">Customer Segments</td></tr>
<tr><td>Key Resources</td><td>Channels</td></tr>
<tr><td colspan="3">Cost Structure</td><td colspan="2">Revenue Streams</td></tr>
</table>

The final business model canvas is founded upon and extended by a cohesive, integrated collection of validated hypotheses about the business model. Hypotheses were iteratively tested against feedback from potential customers, then refined, eliminated, or accepted. The following hypotheses represent all validated conclusions about a potentially viable business model for an automated tailoring company:

Value Proposition

- *Affordability* – Flexible manufacturing would allow customers to tailor their everyday clothing, as well as reducing the price of fitted formal clothing
- *Accommodation* – Inclusive design options would allow customers to order products for a diverse range of personal proportions and unique needs
- *Personalization* – Custom manufacturing would extend inclusive design principles by allowing all customers to design their own unique products
- *Speed* – Advanced automated systems would allow customers to design, order, and receive custom products nearly as fast as generic mass-produced products
- *Accuracy* – Improving order accuracy with advanced measurement technology would greatly reduce costs, logistical complexity, and environmental impacts

Customer Segments

- People who want affordable fitted clothing
- People with unique sensory or other needs for clothing
- People who want or need tailored clothing quickly
- Online clothing retailers attempting to reduce returns and increase satisfaction

Customer Relationships

- A focus on personalization would facilitate close customer relationships
- Measurement and preference profiles would draw repeat customers

Revenue Streams

- Software licensing would not be a viable revenue stream
- The primary revenue stream would be clothing sales to individual customers
- Prices can be higher than mass-manufactured clothing due to added value
- Prices can be low enough to compete with mass manufactured products
- A wide variety of garments for different occasions are viable products
- Measurement services for online retailers would be a viable revenue stream
- In-store revenue streams should be developed through business partnerships

Cost Structure

- Fixed – development of 3D imaging and data processing software
- Fixed – renting and operating manufacturing equipment
- Variable – materials to produce clothing

Channels

- Customers would not be interested in a mall booth sales channel
- Customers would not prefer an independent store location sales channel
- Business should be primarily conducted online to reduce costs
- Multiple online channels should be operated and integrated together
- Internet and social media marketing are ideal for the target market
- Expanding into physical stores is an option but not a priority
- Physical store presence should primarily be through business partners
- Online orders should rely on customers taking their own measurements
- Orders should include a file submission and all selected preferences

Key Activities

- Finding new uses for 3D imaging software would drive business growth
- It would be important to continue refining the 3D imaging technology in use

Key Resources

- Customers would accept and use 3D imaging and measurement tools
- Customers would prefer and adopt smartphone-based 3D imaging tools

Key Partners

- It would be necessary to work with a flexible manufacturing business
- It would be beneficial to work with other clothing brands and businesses
- A social media marketing firm could reach the most potential customers

END – Business Model Deliverable

BEGIN – Discussion

GENERAL RESULTS

Academic Deliverables

The project successfully produced the final deliverables which were proposed and approved during the D.I.S. program and degree application process. In the original document, the goal and deliverables were described as follows:

> Summary
> "The aim of this project is to produce affordable custom-fit clothing, by creating or adapting a 3-D scanning device, 2-D and 3-D modeling software, and a business plan for the commercial sale of such products."
>
> Deliverables
> "This project should result in a business plan for a product (fitted clothing), the product itself, and the software and hardware required to create that product."

It became necessary to adjust the project description and methods from the original proposal, but the final goal and deliverables were maintained and completed.

Academic Results

Beyond the three concrete academic deliverables, this project was developed and approved in expectation of several other academic results. It is expected that the following skills should be developed during the course of a D.I.S. degree and project:

- Long term planning, management, and conflict resolution
- Independent large scale project management

- Passable proficiency in no less than three traditional academic fields
- Unique interdisciplinary knowledge, experience, and proficiency

- Proactive interpersonal networking and coordination
- Academic thesis delivery and defense

Functional Results

Hardware

It was initially expected that a LiDAR or Stereoscopic camera would be necessary to capture accurate 3D images for fitting measurements, and would therefore be required as a deliverable. Multiple cameras and a purchased or constructed mounting system were also considered. However, research and testing identified 3D imaging solutions which can be used with any smartphone camera. Therefore, no unique hardware was required to create the prototype product deliverable.

Software

In place of specialized hardware, the photogrammetry app PolyCam was used to capture, preprocess, and export 3D images. Alternative software was identified but ultimately not required for capturing 3D images, such as RealityCapture.

It was initially expected that existing and new modeling software would be necessary to unwrap a 3D object into 2D cuts of fabric. However, research identified that a data mining pipeline could be used to extract measurements directly from a 3D image, which could then be applied to standard clothing templates. The data mining pipeline successfully extracted all measurements required to produce the prototype product deliverable, and therefore completes the software deliverable.

Prototype Product

It was initially expected that a product would be created personally. However, further consideration identified that the conceptualized automated manufacturing process could be simulated by ordering from a tailor with certain stipulations. A seamstress was contacted and apprised of the project's goals and requirements, then a garment was commissioned with only measurements from the software deliverable. This completes the prototype product deliverable, to be physically presented.

Business Model

The Lean Startup methodology was used to create, test, and refine a business model canvas from the other deliverables and concepts of this project. This method and model were identified as the best representation of the 'business plan' deliverable as it was initially understood and proposed. The canvas is divided into nine categories of design choices, and fully supported by a listed series of hypotheses validated by customer research. This completes the business model deliverable.

GENERAL DISCUSSION

The project results successfully demonstrate that existing technology can be used to automate the tailoring process. At the time of the its proposal, no prior work toward this objective was known, and it was understood to be among the very first attempts to apply 3D imaging and flexible manufacturing technology to the clothing industry.

Prior Work

During the course of the project, businesses independently began to implement partially automated tailoring processes. MTailor produces clothing in the $50-100 price range with an imaging app which locates 16 points on the body using a machine learning algorithm (MTailor 2021). Slo uses an imaging app which collects 90 measurements for jeans with no listed prices (Slo 2023). Unspun uses an iPhone app and LiDAR camera to capture a 3D image, then produce clothing in the $150-250 price range (Unspun 2023).

Although 3D imaging measurement apps increase the regional availability of the tailored products, the products are still offered as premium items for a relatively affluent target audience. Intentionally or not, these three businesses do not make use of automated manufacturing, which could can produce a garment in about four minutes at a cost of about $0.33 (Innovation in Textiles 2017). This limits both production volume and affordability. By contrast, this project aimed to automate the tailoring process end-to-end in order to minimize costs and offer comfortable clothing options to the widest possible variety of people.

Full Automation

To fully automate the tailoring process, this project confirmed the existence and concretely established reputations of flexible automated sewing machines in general, and SoftWear Sewbots in particular (SoftWear Automation n.d.). Documentation of these machines was found alongside discussion of Industry 4.0 in the textile industry (Textile World 2020). While advocating for full automation, it was necessary to consider additional ramifications and alternative ways to reintroduce human elements to the process. Augmented reality 'dressing rooms' were identified as an example of a potentially impactful qualitative feature for an entirely remote ordering system.

Impacts

From the conception of this project, it was understood that a fully automated tailoring process could pose a significant threat to professional tailors. The possibility of this conflict was minimized by focusing on a different target market, but it remains an important consideration for an automation-focused business. If Industry 4.0 technology alters the landscape of the clothing industry too much, the role of tailors in the industry could change dramatically.

For routine fitting work, tailors would not be able to compete with the speed, volume, and efficiency of automated systems. However, their unique expertise in fashion, stylistic adjustments, and aesthetic design could become more valued and desired. When long-standing needs are met, people become able to envision and enjoy something more. After fulfilling needs of inclusion and comfort, it may be both a social responsibility and an opportunity for an automated tailoring business to employ or work with tailors to provide a new, more fulfilling human experience.

Advancing Technology

This project was inspired by newly available 3D printing, data science, and augmented reality technology. It was completed using smart technology which did not exist before the project began. Furthermore, that technology transformed its concept from an affordable walk-in machine for a small business to an app which could be deployed everywhere at once. Similarly, recent advancements in Industry 4.0 technology have enabled unprecedented breakthroughs in many other fields. As investment into Industry 4.0 strategies and systems has dramatically increased over the past few years (IoT Business News 2022), this pattern of accelerated development is likely to continue.

Among Industry 4.0 technologies, the most impactful to date or on the horizon may be AI systems. In less than six months, these have become ubiquitous and transformed a staggering number of social, professional, and philosophical landscapes. AI systems and the technologies used throughout this project act as force multipliers for time and resources which would not otherwise be capable of enacting change. To demonstrate both the potential impact of this project and how quickly it may be surpassed due to the same advancing technology, an experiment was conducted with an AI system.

The experiment attempted to recreate the final business model deliverable with little or no influence from prior knowledge using only the publicly available March 23 version of ChatGPT 3.5 (OpenAI 2023). Eight sentences in five prompts replicated approximately five weeks of informed effort. This would not have the value of a validated business model, but it would be a potent starting point for a lean startup. The following five pages are the user prompts and AI responses from the experiment:

User:
What are some societal and economic implications of industry 4.0 technology? Be curt and direct

ChatGPT:
Some societal and economic implications of Industry 4.0 technology include:

1. Automation of jobs and potential displacement of workers.
2. Increased efficiency and productivity in manufacturing and production processes.
3. Possibility of customization and personalization of products at scale.
4. Development of new business models and revenue streams.
5. Creation of new job roles requiring specialized technical skills.
6. Greater reliance on data and connectivity, raising concerns about privacy and security.
7. Need for investment in infrastructure and upskilling of the workforce to fully realize the potential benefits of Industry 4.0 technology.

User:
Elaborate on points 3 and 7

ChatGPT:
1. Possibility of customization and personalization of products at scale: With Industry 4.0 technology, businesses can use advanced data analytics and automation tools to produce goods tailored to the specific needs and preferences of individual consumers. This can lead to increased customer satisfaction, loyalty, and sales. However, it also requires significant investment in new technologies and processes, and may require businesses to completely rethink their traditional production methods.

2. Need for investment in infrastructure and upskilling of the workforce: Industry 4.0 technology relies heavily on digital connectivity and data management, which means that businesses must invest in infrastructure such as high-speed internet and secure data storage. Additionally, the new technologies and processes associated with Industry 4.0 require specialized technical skills. Therefore, companies need to invest in training and upskilling their workforce to ensure they have the necessary expertise to operate and maintain these technologies. This can also lead to new job opportunities in areas such as data analytics, software development, and robotics.

User:
Contextualize and extend both explanations in the textile and computer peripheral industries. Additionally, highlight use cases and implications of 3D imaging technology in Industry 4.0 systems

ChatGPT:
In the textile industry, Industry 4.0 technology can facilitate the production of customized garments with the help of computer-aided design (CAD) systems, 3D printing, and automated sewing machines. With these tools, textile manufacturers can produce customized clothing and accessories that meet individual customer preferences and specifications. This leads to a more efficient production process, as well as higher customer satisfaction and increased sales. However, this requires significant investment in new technologies and processes, and may require companies to restructure their supply chains and production lines to accommodate these changes.

Similarly, in the computer peripheral industry, Industry 4.0 technology can lead to the creation of highly personalized and customized computer hardware and software products. For example, manufacturers can use advanced data analytics to identify user preferences and tailor their products accordingly. This can improve customer satisfaction and increase sales, but again requires significant investment in new technologies and processes, as well as a shift towards more flexible and adaptable production systems.

3D imaging technology is a key component of Industry 4.0 systems, as it enables the creation of highly accurate and detailed 3D models of objects and environments. In manufacturing, 3D imaging can be used to create virtual prototypes, reducing the need for physical testing and prototyping. This can save time and money, and also enables manufacturers to make design changes more quickly and efficiently. Additionally, 3D imaging can be used for quality control and inspection, ensuring that products meet strict quality standards before they are shipped to customers.

In the healthcare industry, 3D imaging technology can be used to create highly detailed models of human organs and tissues, enabling doctors to diagnose and treat diseases more accurately and effectively. This can lead to improved patient outcomes and reduced healthcare costs over time.

In the entertainment industry, 3D imaging can be used to create highly realistic and immersive virtual environments, enhancing the gaming and entertainment experience for consumers. This can increase engagement and loyalty, as well as create new revenue streams through the sale of virtual goods and experiences.

Overall, the use of 3D imaging technology in Industry 4.0 systems has the potential to revolutionize many industries and transform the way we design, manufacture, and consume goods and services.

User:
Create a list of conceptual business models which can only exist with Industry 4.0 technology, and could not be approximated or created by upgrading traditional technologies or processes

ChatGPT:
Here are some conceptual business models that can only exist with Industry 4.0 technology:

1. On-demand Manufacturing: With Industry 4.0 technology, businesses can use real-time data to optimize their production processes and create highly customized products on-demand. This can lead to reduced inventory costs, faster time-to-market, and improved customer satisfaction.

2. Predictive Maintenance: Industry 4.0 technology allows businesses to collect and analyze data from sensors embedded in their equipment and machinery, enabling them to predict maintenance needs and avoid costly downtime.

3. Asset Performance Management: By using Industry 4.0 technology to monitor and analyze data from their assets, businesses can optimize their performance and reduce maintenance costs over time.

4. Augmented Reality (AR) Commerce: With the help of Industry 4.0 technology, businesses can create highly personalized AR shopping experiences that enable customers to visualize products in their own environment before making a purchase.

5. Smart Supply Chains: Industry 4.0 technology allows businesses to optimize their supply chains by using real-time data to track inventory, anticipate demand, and optimize shipping and logistics.

6. Digital Twinning: With the help of Industry 4.0 technology, businesses can create digital twins of their physical assets, enabling them to test and optimize their performance in a virtual environment before making changes in the physical world.

7. Blockchain-enabled Transactions: Industry 4.0 technology can enable businesses to use blockchain technology to create secure, transparent transactions and track the movement of goods and assets throughout their supply chain.

These business models are made possible only by the advanced data analytics, connectivity, and automation tools offered by Industry 4.0 technology, and could not be replicated or approximated by upgrading traditional technologies or processes.

User:
Create a full but concise business model canvas for a clothing startup which implements the first concept by automating the tailoring process. Assume that customers use a smartphone and low-skill 3D imaging software to take all measurements, then order online.

ChatGPT:
Sure, here's a concise Business Model Canvas for a clothing startup that automates the tailoring process using Industry 4.0 technology:

Key Partners:
- Technology providers for 3D imaging software, automated sewing machines, and data analytics tools.
- Fabric and material suppliers.
- Logistics and delivery partners.

Key Activities:
- Developing and implementing 3D imaging software to enable customers to take their measurements using their smartphone.
- Implementing automated sewing machines and robotics to produce customized garments.
- Integrating data analytics tools to optimize production processes and inventory management.
- Marketing and advertising the service to attract new customers.

Key Resources:
- Automated sewing machines, robotics, and 3D imaging software.
- Skilled technical personnel to operate and maintain the technology.
- Access to high-quality fabric and materials.
- Marketing and branding materials to promote the service.

Value Proposition:
- Highly personalized and customized clothing tailored to individual customer preferences and specifications.
- Fast delivery times and reduced lead times.
- Lower costs and increased efficiency compared to traditional tailoring methods.

Customer Segments:
- Young and tech-savvy individuals who value customization and personalization in their clothing.
- Busy professionals who don't have time for traditional tailoring methods.
- Athletes and individuals with unique body shapes who struggle to find well-fitting clothing in traditional stores.

Channels:

- Online platform for customers to take their measurements and place orders.
- Social media advertising and influencer marketing to reach target customer segments.
- Partnerships with fashion bloggers and influencers to promote the service.

Customer Relationships:

- Strong customer support and personalized communication throughout the ordering and delivery process.
- Ability to provide updates and tracking information throughout the production and delivery process.

Revenue Streams:

- Revenue generated from the sale of customized clothing and accessories.
- Additional revenue streams from accessories such as belts, ties, and scarves.
- Potentially, additional revenue from licensing of technology to other clothing manufacturers.

Cost Structure:

- Costs associated with technology acquisition and implementation.
- Costs associated with sourcing high-quality fabric and materials.
- Salaries and wages for technical personnel and customer support staff.
- Marketing and advertising costs to promote the service.

Overall, this clothing startup will use Industry 4.0 technology to provide a highly personalized and efficient tailoring service, leveraging automation, data analytics, and 3D imaging software to deliver custom clothing quickly and at a lower cost than traditional tailoring methods.

PERSONAL RESULTS

Disciplines

Originally, I defined my program of study to achieve two primary goals: to resolve a quandary of choice between four fields of interest, and thereby prepare me for this project. I very much wanted to learn certain subjects from each field, but I knew that I would struggle with other required content which I could not find interesting or useful. The project and curriculum were developed in tandem to accomplish something which mattered to me, by way of learning concepts and skills which I was passionate about. I condensed my four fields into three disciplines, then completed non-elective courses comparable to traditional majors in each discipline. To my surprise, I learned much more.

Synthesis

Over time, I discovered many potent intersections and interactions between the fields of computer science, industrial engineering, psychology, and business. The first intersection I recognized at a surface level was the discipline I studied, of human factors engineering, between industrial engineering and psychology.

I quickly but ever-increasingly learned the value of psychology in each other field for self-monitoring and design. Spreadsheets from business proved to nicely complement data mining techniques from computer science for different types of data analysis. Psychology provided a robust understanding of memory and other human factors which industrial engineering used a brief approximation of for interaction design. Industrial engineering provided tools to evaluate the usability of software tools and applications from computer science. Business contextualized the resources and processes necessary for computer science to create products. Psychology explained the merits and pitfalls of business management and marketing strategies. Computer science facilitated the habits of systematic analysis to understand concepts of behavioral psychology.

These go on and on, yet still only represent relatively simple and easily summarized second-order insights. In brief, at the synthesis of my disciplines, I learned the immense value of simultaneously understanding diverse perspectives.

Resume

The following document is my current resume, primarily developed from this project:

Ian Off

Interdisciplinary Studies Engineering

(720) 550-2965
atimeofday@gmail.com
Bozeman, MT 59718

EDUCATION

Bachelor of Science – Interdisciplinary Studies Engineering — **MSU Bozeman**
August 2019 – May 2023 — **3.92 GPA**

- Disciplines: Computer Science, Industrial Engineering, Business, Psychology
- Focus areas: Industry 4.0, Human Factors Engineering
- Interdisciplinary capstone project: Industry 4.0 Tailoring – see Projects below

Associate's Degree – General Studies — Red Rocks Community College
August 2018 – May 2019 — 4.00 GPA

Diploma – General Studies — Bear Creek High School
August 2015 – May 2019 — 4.17 GPA

PROJECTS

DIS Capstone – Industry 4.0 Tailoring
January 2022 – May 2023

I developed software to take measurements from 3D images, prototyped an automated apparel-tailoring process, and developed a business model around it using the Lean Startup process.

Integrated database management system – SQLite, Java
August 2021 – December 2021

Little man stack machine emulator – Assembly, C, C++, Java, littlemanstackmachine.org
January 2023 – May 2023

Linux – Learning, using, extending, and contributing
February 2022 – Ongoing

SKILLS

Computer Science
- Java, SQLite, C++, C, Python
- Debugging, data mining, algorithms
- Linux, OCI, CLI, git

Business
- Lean startup methodology
- Financial analysis, accounting
- Organizational management

Human Factors Engineering
- Project & team management
- Usability, inclusive design
- System design & analysis

Personal
- Communication, coordination
- Digital fluency & learning
- Initiative, independence

EXPERIENCE

VR Consultant — MSU Bozeman, Digital Library Initiatives
January 2020 – May 2023 — Part Time

CONCLUSION

This project was successful. All requirements were completed in full and on time.

Primary Objective

The project results demonstrate that a small business could produce affordable mass-customized clothing with a photogrammetry app, a data mining pipeline, and existing flexible manufacturing machinery. This would make it possible for a much wider variety of people to find and purchase clothing for their size, shape, and unique needs.

Future Work

Together, the deliverables of this project constitute a successful proof of concept for a large commercial system to automate the tailoring process. However, they are still only prototypes completed by a single person with minimal resources. Much work remains to refine and integrate those prototypes, then deploy a completed system.

The first objective of future work should be to generalize and optimize the data mining pipeline, in order to filter out environment data and noise from any 3D image of one type of thing. This is a significant area of weakness for the prototype pipeline and final software deliverable from this project. The prototype pipeline and final software deliverable, although designed to minimize the influence of human knowledge and guidance, were not tested on alternative inputs due to time and resource constraints.

The second objective of future work should be to update, finalize, and enact the business model. This project is meant to be one step toward the long term goal of providing affordable comfortable clothing for as many different people as possible. To achieve this, no amount of technical or theoretical refinement can substitute for industrial deployment.

Industry 4.0

This project applies Industry 4.0 technology and human factors engineering to address a previously intractable flaw of the clothing industry, and introduce inclusive design. While impactful, this is not an isolated event. Advancements among smart factories, integrated information systems, data analysis software, AI, and similar technologies continue to accelerate, promising even greater upheavals throughout other industries. For many fields and endeavors untouched by automation, revolution is on the horizon.

END – Discussion

BEGIN – References

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